Self-purging bicycle tube valve

Abstract

A self-purging valve (simulated abstract for the replay test).

Detailed Description

Title: valve core for a tyre valve, seat ring, and methods of use and manufacture

Technical field

[0001] The present disclosure relates to tyre valves. It relates in particular to a valve core for a tyre valve that admits air into a tyre chamber containing sealant, to a tyre valve and a tyre chamber fitted with such a valve core, to a seat ring for such a valve core, to a method of cleaning a tyre valve during an inflation cycle, and to methods of making the seat ring and of assembling the valve core. The tyre chamber of a bicycle is described by way of example.

Background

[0002] A liquid sealant is commonly introduced into the tyre chamber of a bicycle, whether a bicycle inner tube or a tubeless tyre and rim cavity, to seal punctures. The sealant is typically a latex emulsion containing ammonia and suspended particles, and it is introduced through the valve stem of the tyre valve. On each inflation a small quantity of sealant is drawn into the valve core by the pressure surge and by capillary action along the stem of the poppet. Between rides the sealant dries, and a sealant film forms on surfaces of the valve core.

[0003] A simulated replacement paragraph.

[0004] US11179979B2 describes a valve stem system in which a first valve is arranged at the end of the valve stem body that lies in the volume enclosed by the tyre and the rim. The first valve is described as a self-sealing one-way valve that is operable although a sealant is present, and the document states that the first valve usually prevents sealant from entering the valve stem.

[0005] US11148485B2 describes a valve stem assembly in which a plunger of the valve may be used as a tool against clogging when sealant accumulates inside the body of the valve stem.

[0006] US20210323363A1 states that valves used with tubeless bicycle tyres are often clogged by sealant due to their small diameter. It describes a valve having a larger internal diameter than typical bicycle valves, which is said to result in higher air flow and reduced clogging.

[0007] A simulated inserted paragraph answering an enablement gap.

[0008] A second one.

[0009] US2019/0123456A1 describes a valve core having a bore that is enlarged to reduce capillary rise of sealant along the stem of the poppet, a sealing face coated with a fluoropolymer layer that is described as reducing adhesion of cured sealant, a rigid seat, and a removable insert that captures deposits of sealant and can be pulled out and cleaned or replaced without removal of the whole valve core.

[0010] Valves in which further travel of a poppet opens a further flow path are described in other fields. US5918628A describes a check valve having a spring-biased poppet with a first stage opening and second stage openings spaced axially along the poppet. Flow through the second stage openings is restricted when the poppet first moves from its closed position, and increases after a predetermined axial movement of the poppet. US4044791A describes a valve assembly having a floating valve seat that is yieldably biased into engagement with a poppet. After the floating valve seat meets an abutment, the fluid unseats the poppet from the floating valve seat and flows through secondary passageways in the poppet.

[0011] Valves in which a flow of fluid cleans a seat, or deforms a ring, are also described in other fields. US3570510A describes a valve device in which a clean fluid such as gas or air is supplied to the valve seat from a circular space through an annular slit defined between two concentric seat rings, so as to keep the valve seat clean. US3595265A describes a check valve having a groove in the valve face that acts as an orifice when the valve is closed; when the valve opens, the stream of fluid flowing past the valve face and the groove flushes away foreign particles lodged there. US5924444A describes a valve in which, in the fully opened position, fluid leaving an orifice radially deforms an elastomeric ring outwardly and forms a gap between a side wall and the elastomeric ring, the gap enhancing the egress of fluid, and the elastomeric ring returns to its non-deformed shape as the valve closes.

Technical problem

[0012] The technical problem addressed is the removal, on each inflation cycle and without a tool, of a sealant film that has already formed on surfaces of the valve core.

[0013] Air that enters the tyre chamber through the main passage flows between the sealing face of the poppet and the sealing edge. A longer travel of the poppet along the main passage, or a main passage of larger section, enlarges that same flow path. The stream of air remains within the main passage, and a surface of the valve core that lies outside the main passage, for example an outer surface of the seat ring, is not brought into the stream by either measure.

[0014] In the conventional valve cores of the bench test reported under "Examples and data", the air admitted on every inflation cycle passed through the main passage, and those valve cores nevertheless reached their first failure within eighteen inflation cycles.

[0015] The valve core described herein has a purge gap, which is a secondary flow path distinct from the main passage, and which is opened by a load that the poppet applies to a radially deflectable element when the poppet travels beyond its normal stroke. In the documents referred to above, further travel of a poppet increases the flow through the valve, and the gap formed at the elastomeric ring enhances the egress of fluid. The purge gap described herein instead directs air, on its way into the tyre chamber, across a surface of the valve core on which sealant film accumulates.

Summary

[0016] According to a first aspect there is provided a valve core for a tyre valve, the valve core comprising a core body, a radially deflectable element, and a poppet that is movable along an axis of the valve core. The poppet has a sealing face that closes against a sealing edge at rest. The poppet is movable away from the sealing edge through a normal stroke, in which a main passage is open between the sealing face and the sealing edge, and is movable beyond the normal stroke through an over-travel. A load-applying surface that is moved by the poppet is spaced from the radially deflectable element throughout the normal stroke and bears on the radially deflectable element in the over-travel. The axial load so applied produces a radial deflection of the radially deflectable element, and the radial deflection opens a purge gap. The purge gap is a secondary flow path that is distinct from the main passage and that is closed at rest and throughout the normal stroke. When the axial load is removed the radially deflectable element undergoes shape recovery and the purge gap closes.

[0017] Two features of the first aspect act together. The first is that the normal stroke and the over-travel are two travel ranges of the same poppet, and the load-applying surface bears on the radially deflectable element only in the second of them, so that the radially deflectable element is not deflected by the load-applying surface during ordinary opening of the tyre valve. The second is that the purge gap is a flow path of its own. The purge gap does not run between the sealing face and the sealing edge, and a main passage that is enlarged, by whatever cause, is not a purge gap.

[0018] The purge gap is opened by the mechanical load of the poppet. The radial deflection is produced by the load-applying surface bearing on the radially deflectable element, and not by a flow of fluid acting on the radially deflectable element. When the purge gap is open, air at pump pressure flows through the purge gap, across a surface of the valve core on which sealant film accumulates, and into the tyre chamber. This flow is the air purge.

[0019] In embodiments the radially deflectable element is a seat ring that carries the sealing edge, the load-applying surface is a purge shoulder on the poppet, the radial deflection is a radial expansion of the seat ring, and the purge gap is bounded by an outer surface of the seat ring and by a bore surrounding the seat ring.

[0020] According to a second aspect there is provided a tyre valve comprising a valve stem and a valve core according to the first aspect. In some embodiments the valve core is a removable threaded valve core received in the valve stem. There is also provided a tyre chamber fitted with the tyre valve, for example a bicycle inner tube or a tubeless tyre and rim cavity, which in use contains sealant.

[0021] In some embodiments of the second aspect the valve core is integral with the valve stem.

[0022] According to a third aspect there is provided a seat ring for a valve core of a tyre valve. The seat ring is tubular and comprises a sealing edge, on which a sealing face of a poppet of the valve core seats, and a wall having a load-converting geometry. The load-converting geometry is such that an axial load, applied to the seat ring by a load-applying surface that is moved by the poppet in an over-travel of the poppet beyond a normal stroke, produces a radial expansion of the wall that opens a purge gap distinct from a main passage of the valve core. The seat ring undergoes shape recovery, which closes the purge gap, when the axial load is removed. The seat ring may be supplied as a replacement part for a valve core, and the valve core may be supplied as a replacement part for a tyre valve.

[0023] According to a fourth aspect there is provided a method of cleaning a tyre valve during an inflation cycle, the tyre valve having a valve core according to the first aspect and being fitted to a tyre chamber that contains sealant. The method comprises coupling a pump head to the tyre valve; applying with the pump head a pressure differential that moves the poppet through the normal stroke; moving the poppet into the over-travel, so that the load-applying surface bears on the radially deflectable element and the purge gap opens; passing air through the open purge gap into the tyre chamber; and removing the pump head, whereupon the pressure differential collapses, the poppet returns, the radially deflectable element undergoes shape recovery and the purge gap closes. The method is performed with the valve core in place in the tyre valve.

[0024] According to a fifth aspect there is provided a method of making a seat ring according to the third aspect, the method comprising laser cutting a tube of NiTi alloy, shape-setting the cut tube in a fixture that holds the load-converting geometry, and quenching. In some embodiments the method further comprises electropolishing. There is also provided a method of assembling a valve core, comprising press-fitting the seat ring into a core body, against a locating shoulder of the core body, with an interference fit.

[0025] In some embodiments the valve core comprises a return bias that urges the poppet towards the sealing edge and that has a two-rate characteristic, with a first spring stage that is overcome at a crack pressure and a second spring stage that is overcome at an over-travel differential greater than the crack pressure. The poppet then moves through the normal stroke at a pressure differential between the crack pressure and the over-travel differential, and reaches the over-travel only when the higher pressure differential is applied. In embodiments the return bias is a return spring.

[0026] In some embodiments the purge gap is annular and lies between an outer surface of the seat ring and a bore surrounding the seat ring, the bore being the core body bore or a bore of the valve stem. The air purge then passes over the outer surface of the seat ring, which lies outside the main passage.

[0027] In some embodiments the purge gap runs through the wall of the seat ring, so that the outer surface of the seat ring can be retained along its whole length.

[0028] In some embodiments the load-applying surface is a purge shoulder on the poppet, below the sealing face, that is spaced from the seat ring throughout the normal stroke. The axial load is then applied by the poppet itself, and the position of the poppet at which the purge shoulder meets the seat ring marks the end of the normal stroke.

[0029] In some embodiments the load-converting geometry is a convoluted wall having at least one convolution. Under the axial load the shape of the convolution changes by bending of the convoluted wall, and the change of shape produces the radial expansion.

[0030] In some embodiments the seat ring is made of a NiTi alloy that is superelastic and that has an austenite finish temperature lower than 5 °C, so that the seat ring is superelastic at riding temperatures.

[0031] In some embodiments the shape recovery of the seat ring produces a snap-back wipe, in which the seat ring moves across the sealing face of the poppet as the seat ring recovers. The snap-back wipe is a cleaning action additional to the air purge and is not present in every embodiment.

[0032] In some embodiments the seat ring is a polymer seat ring, for example of PEEK, which has radial compliance and which undergoes shape recovery more slowly and with lower force than a seat ring of superelastic NiTi alloy.

[0033] In some embodiments the valve core has a two-piece seat, in which a rigid seat provides the sealing edge and a separate wiper ring below the rigid seat is the radially deflectable element on which the purge shoulder bears. Sealing then takes place on an element that is not deflected in the over-travel.

[0034] In some embodiments the core body is made of brass. In other embodiments the core body is made of stainless steel, for resistance to corrosion where the sealant contains ammonia. In some embodiments the sealing face is of hard-anodised aluminium, which gives the sealing face a hard surface.

[0035] In some embodiments the dimensions of the seat ring, the over-travel distance, the purge gap width, the crack pressure and the over-travel differential have the values and ranges set out under "Parameters and ranges".

[0036] A thermal variant, in which the seat ring opens the purge gap on warming, is described under "Materials and shape recovery".

[0037] Features described in connection with one aspect apply to the other aspects, and the optional features are combinable with one another as set out under "Combinations".

Brief description of the drawings

[0038] Embodiments are described below, by way of example only, with reference to the accompanying drawings. The drawings are schematic and are not to scale, and like reference numerals denote like parts.

[0039] FIG. 1 is an axial section of a tyre valve at rest, with the purge gap closed.

[0040] FIG. 2 is the axial section of FIG. 1 with the poppet in over-travel and the purge gap open, with arrows showing the flow of air through the main passage and through the purge gap.

[0041] FIG. 3 is an exploded perspective view of the valve core of FIG. 1.

[0042] FIGS. 4A and 4B are detail sections of the seat ring having a convoluted wall with one convolution and with two convolutions respectively, showing a fit band and a free portion of the seat ring.

[0043] FIG. 5 is a qualitative force–travel diagram of the two-rate characteristic.

[0044] FIG. 6 is a sequence of four panels showing the use of the tyre valve over one inflation cycle.

[0045] FIG. 7 is a flow diagram of a manufacturing sequence of the seat ring and of press-fitting.

[0046] FIG. 8 is an axial section of an embodiment having a polymer seat ring.

[0047] FIGS. 9A and 9B are axial sections, at rest and in over-travel respectively, of an embodiment having a two-piece seat with a rigid seat and a wiper ring.

[0048] FIG. 10 is a schematic axial section of the thermal variant.

[0049] FIGS. 11A, 11B and 11C are axial sections of embodiments having a slotted sleeve, a split seat ring and a tapered seat ring respectively.

[0050] FIGS. 12A and 12B are axial sections of embodiments having, respectively, a relieved core body bore and a purge flow through the wall of the seat ring.

[0051] FIG. 13B is an axial section of an embodiment in which the valve core is integral with the valve stem.

Definitions and interpretation

[0052] The following definitions apply throughout this description and in the claims, unless the context requires otherwise.

General construction

[0053] "Comprising" and its variants are open, non-exclusive terms. A component, step or combination described as comprising stated elements may include further elements that are not stated. Where a closed or partly closed meaning is intended, "consisting of" or "consisting essentially of" is used expressly.

[0054] Language such as "in some embodiments", "in embodiments", "optionally", "for example" or "may" identifies a feature as optional or preferred, and not as essential to every embodiment. Where a feature is described without such language in the context of one embodiment, the feature is presented as part of that embodiment, and is not thereby made essential to any other embodiment.

[0055] "A" and "an" mean one or more, unless the context requires a single instance. For example, "a convolution" covers an embodiment having one convolution and an embodiment having two convolutions.

[0056] Terms that describe a relationship between components, for example "on", "bears on", "contacts", "meets", "coupled to", "carried by" and "retained in", cover a direct relationship and a relationship in which a further part lies between the components, unless the relationship is stated to be direct.

[0057] A numerical range includes both of its stated endpoints and every value and every sub-range within the range, including each intermediate range and preferred range stated for the same parameter. A range that is stated by an upper bound only is open at its lower end.

[0058] Where a value is qualified by "about" or "approximately", the value covers the ordinary manufacturing tolerance and measurement tolerance for the parameter concerned.

[0059] Steps of a method are not limited to the order in which the steps are presented, unless the context requires an order. Where a physical relationship fixes an order, that relationship governs: the normal stroke of the poppet precedes the over-travel, and the shape recovery follows removal of the axial load applied in the over-travel.

Component and structural terms

[0060] As used herein, **"seat ring"** means a tubular element, retained in the core body and having radial compliance, that has a sealing edge against which the sealing face of the poppet closes at rest, and a wall that converts an axial load applied by the purge shoulder in the over-travel into a radial expansion that opens the purge gap, the seat ring undergoing shape recovery when the axial load is removed. The seat ring is not limited to a material. A seat ring of superelastic NiTi alloy is preferred, and a polymer seat ring is another example.

[0061] As used herein, **"sealing edge"** means the surface on which the sealing face of the poppet seats to close the tyre valve at rest. In embodiments having a seat ring the sealing edge is an edge of the seat ring. In the two-piece seat the sealing edge is provided by the rigid seat.

[0062] As used herein, **"core body bore"** means the internal surface of the core body that faces the outer surface of the seat ring. In embodiments the core body bore is one boundary of the purge gap.

[0063] As used herein, **"convoluted wall"** means a wall of the seat ring that is not straight in section, so that an axial load on the seat ring translates into a radial expansion of the convoluted wall. A convoluted wall having one convolution and a convoluted wall having two convolutions are both within this definition.

[0064] As used herein, **"convolution"** means a single wave in the section of a convoluted wall.

[0065] As used herein, **"purge shoulder"** means a surface of the poppet, below the sealing face, that is spaced from the seat ring throughout the normal stroke, that contacts the seat ring only in the over-travel, and that applies the axial load which produces the radial expansion. The purge shoulder is the preferred load-applying surface.

[0066] As used herein, **"main passage"** means the flow path by which air flows into the tyre chamber past the sealing edge when the sealing face of the poppet is lifted off the sealing edge. A flow path that runs between the sealing face and the sealing edge is part of the main passage, whatever the size of that flow path, and whether or not that flow path is enlarged by a deflection of the element on which the poppet seats.

[0067] As used herein, **"purge gap"** means a secondary flow path into the tyre chamber that is distinct from the main passage, that is closed at rest and throughout the normal stroke, and that is opened in the over-travel by the deflection that the load-applying surface produces in the radially deflectable element. The purge gap is distinguished from the main passage by where the purge gap runs and by the surfaces that the purge gap sweeps, and not by its size: the purge gap does not run between the sealing face and the sealing edge, and the purge gap is bounded by a surface of the radially deflectable element that lies outside the main passage. In the over-travel, air flows into the tyre chamber through the purge gap in addition to the main passage. In the preferred form the purge gap is an annular clearance between the outer surface of the seat ring and a bore surrounding the seat ring, the bore being the core body bore, which is preferred, or a bore of the valve stem.

[0068] In some embodiments the purge gap is a flow path through the wall of the seat ring, or is a flow path at one or more locations around the seat ring that does not extend around the whole circumference, as described under "Purge gap arrangements and retention of the seat ring".

[0069] As used herein, the purge gap is **"closed at rest"** when the surfaces that bound the purge gap are in the position that those surfaces take when no load is applied to the radially deflectable element by the load-applying surface. The expression does not require closure that is gas-tight to a stated degree; the tyre valve is closed at rest by the sealing face of the poppet on the sealing edge.

Threshold, pressure and travel terms

[0070] As used herein, **"pressure differential"** means the difference between the pressure upstream of the poppet, applied by a pump head, and the pressure in the tyre chamber.

[0071] As used herein, **"crack pressure"** means the pressure differential across the poppet at which the sealing face of the poppet first lifts off the sealing edge. In embodiments having a return spring with a two-rate characteristic, the crack pressure is the pressure differential that overcomes the first spring stage.

[0072] As used herein, **"over-travel differential"** means the pressure differential across the poppet, greater than the crack pressure, that is exceeded for the poppet to reach the over-travel. In embodiments having a return spring with a two-rate characteristic, the over-travel differential is the pressure differential that overcomes the second spring stage.

[0073] As used herein, **"normal stroke"** means the range of travel of the poppet that begins when the sealing face lifts off the sealing edge and that ends at the position of the poppet at which the load-applying surface first meets the radially deflectable element. Throughout the normal stroke the load-applying surface is spaced from the radially deflectable element. In embodiments in which the poppet is moved by the pressure differential, the poppet is in the normal stroke when the pressure differential is between the crack pressure and the over-travel differential.

[0074] As used herein, **"over-travel"** means the range of travel of the same poppet beyond the normal stroke, in which, and only in which, the load-applying surface bears on the radially deflectable element. The normal stroke and the over-travel are two ranges of travel of one poppet, and the position at which the load-applying surface first meets the radially deflectable element is the boundary between them. In the embodiments described the poppet reaches the over-travel when the pressure differential exceeds the over-travel differential, which is the preferred actuation.

[0075] An element that carries the sealing edge is loaded by the sealing face of the poppet at rest. A statement herein that the radially deflectable element is loaded only in the over-travel refers to the axial load applied by the

load-applying surface.

[0076] As used herein, **"over-travel distance"** means the distance that the poppet moves beyond the normal stroke.

[0077] As used herein, **"two-rate characteristic"** means a force–travel behaviour of the return bias that has a lower rate through the normal stroke and a higher rate that is overcome for the poppet to reach the over-travel. Constructions that give a two-rate characteristic are described under "Return bias and load-applying surface".

[0078] As used herein, **"first spring stage"** means the part of the two-rate characteristic that the crack pressure overcomes during the normal stroke, and **"second spring stage"** means the part of the two-rate characteristic that the over-travel differential overcomes to produce the over-travel.

Material and property terms

[0079] As used herein, **"superelastic"** means the property of a material that recovers a large strain on unloading at a temperature above the austenite finish temperature of the material, as distinct from a thermally triggered response in which a change of temperature produces the recovery. A material that recovers its shape on unloading by another mechanism is not superelastic as the term is used herein, and is within the definition of shape recovery given below.

[0080] As used herein, **"NiTi alloy"** means an alloy based on nickel and titanium. Examples are given under "Materials and shape recovery".

[0081] As used herein, **"austenite finish temperature"** means the temperature above which an alloy such as NiTi alloy is fully austenitic and is superelastic.

[0082] As used herein, **"martensite start temperature"** means the temperature below which an alloy such as NiTi alloy begins to transform to martensite and correspondingly loses superelastic recovery.

[0083] As used herein, **"radial expansion"** means an outward movement of the wall of the seat ring, or of a wall of another radially deflectable element, under an axial load, that opens the purge gap. The term does not require that the wall stretches uniformly around its circumference. In embodiments described herein the radial expansion arises from a change, under the axial load, in the shape of a load-converting geometry such as a convoluted wall.

[0084] As used herein, **"radial compliance"** means the capacity of the radially deflectable element to expand radially under the axial load applied by the load-applying surface and to return substantially to its radial dimension at rest when the axial load is removed.

[0085] As used herein, **"shape recovery"** means the return of the radially deflectable element towards its shape at rest when the axial load applied by the load-applying surface is removed, which closes the purge gap. The term is not limited to a mechanism of recovery. The shape recovery is superelastic in a seat ring of superelastic NiTi alloy, and is elastic recovery of another kind in other embodiments, for example in a polymer seat ring. The speed and the force of the shape recovery differ between embodiments.

Cleaning-action terms

[0086] As used herein, **"air purge"** means a first cleaning action, being the flow of air at pump pressure through the open purge gap, across a surface of the valve core on which sealant film accumulates, and into the tyre chamber.

[0087] As used herein, **"snap-back wipe"** means a second, mechanical cleaning action that is present in some embodiments, in which the seat ring, on shape recovery, moves across the sealing face of the poppet. In the preferred embodiment the snap-back wipe is produced by the superelastic recovery of a seat ring of NiTi alloy. An embodiment in which the shape recovery is slower or weaker, for example an embodiment having a polymer seat ring, provides the air purge whether or not that embodiment also provides a snap-back wipe of comparable effect.

Fluid, chamber and cycle terms

[0088] As used herein, **"sealant"** means a liquid, for example a latex emulsion containing ammonia and suspended particles, that is introduced into the tyre chamber through the tyre valve to seal punctures.

[0089] As used herein, **"sealant film"** means a layer of sealant, drawn into the valve core, that dries and cures on a surface of the valve core between uses. The term covers a sealant film that is partially cured and a sealant film that is cured.

[0090] As used herein, **"tyre chamber"** means the pressurised volume into which the tyre valve admits air, for example a bicycle inner tube or a tubeless tyre and rim cavity.

[0091] As used herein, **"pump head"** means a device that is coupled to the tyre valve and that applies the pressure differential, for example the head of a floor pump.

[0092] As used herein, **"inflation cycle"** means one application and removal of the pressure differential by a pump head.

Alternative-embodiment terms

[0093] As used herein, **"polymer seat ring"** means a seat ring made of a polymer that has radial compliance, for example PEEK.

[0094] As used herein, **"two-piece seat"** means an arrangement in which the sealing edge is provided by a rigid seat, and a separate wiper ring below the rigid seat is the radially deflectable element on which the purge shoulder bears in the over-travel.

[0095] As used herein, **"rigid seat"** means, in the two-piece seat, the element that provides the sealing edge.

[0096] As used herein, **"wiper ring"** means, in the two-piece seat, a thin ring having radial compliance that is positioned below the rigid seat, that is deflected outwards by the purge shoulder in the over-travel, and that undergoes shape recovery and wipes when the axial load is removed.

[0097] As used herein, **"thermal variant"** means an embodiment in which the seat ring is of a NiTi alloy trained with two-way shape memory and has an austenite finish temperature of approximately 35 °C to 40 °C, so that warming of the seat ring, and not the over-travel of the poppet, opens the purge gap.

Dimensional and process terms

[0098] As used herein, **"convolution amplitude"** means the height of a convolution of the convoluted wall, measured from the mean line of the convoluted wall.

[0099] As used herein, **"purge gap width"** means the radial size of the purge gap when the purge gap is fully open.

[0100] As used herein, **"shape-setting"** means a heat treatment that is applied to a seat ring held in a fixture, and that fixes in the material of the seat ring, for example NiTi alloy, the convoluted wall or other load-converting geometry held by the fixture.

Genus terms

[0101] As used herein, **"radially deflectable element"** means an element of the valve core that is deflected radially by the axial load which the load-applying surface applies in the over-travel, that thereby opens the purge gap, and that undergoes shape recovery when the axial load is removed. The term covers the seat ring, including the seat ring of superelastic NiTi alloy and the polymer seat ring, and covers the wiper ring of the two-piece seat. The expression "radially compliant seat element" has the same meaning.

[0102] In some embodiments the radially deflectable element is a slotted sleeve, a split seat ring or a tapered seat ring, as described under "Forms of the radially deflectable element".

[0103] As used herein, **"return bias"** means an element, or a combination of elements, that urges the poppet towards the sealing edge. The return spring is the preferred return bias.

[0104] As used herein, **"load-applying surface"** means a surface that is moved by the poppet, that is spaced from the radially deflectable element throughout the normal stroke, and that bears on the radially deflectable element in the over-travel. The purge shoulder on the poppet is the preferred load-applying surface.

[0105] In some embodiments the load-applying surface is a surface of a separate collar carried by the poppet, is conical, or is a surface of a member through which the return spring acts, as described under "Return bias and load-applying surface".

[0106] As used herein, **"load-converting geometry"** means a shape of the radially deflectable element by which an axial load on the radially deflectable element is converted into a radial deflection. The convoluted wall is the preferred load-converting geometry. Other forms are described under "Forms of the radially deflectable element".

[0107] As used herein, **"secondary flow path"** means a flow path into the tyre chamber other than the main passage. The purge gap is a secondary flow path that is closed at rest and that is opened in the over-travel.

[0108] A term that is defined by the function performed, including "radially deflectable element", "return bias", "load-applying surface", "load-converting geometry" and "secondary flow path", covers each structure described herein that performs the function, and is not limited to the preferred structure, unless the passage or the claim concerned recites the preferred structure expressly.

Principal embodiment: structure

[0109] A principal embodiment of the tyre valve 1 is described with reference to FIGS. 1 to 3 and FIG. 5. FIG. 1 is an axial section of the tyre valve 1 at rest. FIG. 2 is the same section with the poppet 40 in the over-travel 116. FIG. 3 is an exploded perspective view of the valve core 5. FIG. 5 is a qualitative force–travel diagram. The valve cores tested in Example 1, under "Examples and data", are valve cores according to the principal embodiment.

[0110] The tyre valve 1 comprises a valve stem 10 and a valve core 5. The valve stem 10 is a tubular member that is fixed to a tyre chamber 80 and has a bore that receives the valve core 5. The tyre valve 1 admits air into the tyre chamber 80 and retains it there. The tyre chamber 80 is indicated in FIG. 1 by a boundary in broken lines and contains sealant in use.

[0111] A pump head 90 is a device that is coupled to the tyre valve 1 and applies a pressure differential across the poppet 40. The pressure differential is the difference between the pressure upstream of the poppet 40 and the pressure in the tyre chamber 80. The pump head 90 is shown in broken outline in FIG. 1, where it is not engaged, and in solid outline in FIG. 2, where it is engaged on the valve stem 10. The pump head 90 is a conventional external device and is not a part of the valve core 5.

[0112] In the principal embodiment the valve core 5 is an assembly of a core body 20, a seat ring 30, a poppet 40 and a return spring 50. In some embodiments the valve core 5 also carries a lock nut 70. FIG. 3 shows these parts separated along the valve axis in their order of assembly: the core body 20, the seat ring 30, the poppet 40 with the return spring 50 concentric on the stem of the poppet 40, and the lock nut 70.

[0113] The core body 20 is fixed in the valve stem 10. In the principal embodiment the core body 20 has a thread by which it is screwed into the bore of the valve stem 10. The core body 20 carries the seat ring 30 and guides the poppet 40 along the valve axis. The core body 20 has a core body bore 22, which is the internal surface of the core body 20 that faces the outer surface of the seat ring 30. The core body 20 also has a locating shoulder 24, which is an internal step of the core body 20 against which the seat ring 30 is seated axially. Materials of the core body 20 are described under "Materials and shape recovery".

[0114] The seat ring 30 is a short tubular element. An end edge of the seat ring 30, also referred to as its upper edge, forms a sealing edge 32 on which the sealing face 42 of the poppet 40 seats to close the tyre valve 1. The seat ring 30 is pressed into the core body 20 with a light interference fit and is located against the locating shoulder 24.

[0115] The seat ring 30 has a convoluted wall 34, that is, a wall that is not straight in section. The convoluted wall 34 has at least one convolution 36, and each convolution 36 is shallow. In some embodiments the convoluted wall 34 has one convolution 36, as shown in FIG. 1. In other embodiments the convoluted wall 34 has two

convolutions. The convoluted wall 34 has the effect that an axial load applied to the seat ring 30, for example at its upper edge, translates into radial expansion of the seat ring 30.

[0116] In the principal embodiment the seat ring 30 is a single part that both carries the sealing edge 32 and undergoes the radial expansion. The closing of the tyre valve 1 and the opening of the purge gap 60 are thereby provided by one part, without the additional part and the additional axial length of the two-piece seat described under "Forms of the radially deflectable element".

[0117] The seat ring 30 is an example of a radially deflectable element. Other forms of the radially deflectable element are described under "Forms of the radially deflectable element". Materials of the seat ring 30, including the superelastic NiTi alloy of the valve cores of Example 1, are described under "Materials and shape recovery".

[0118] The poppet 40 is a member that is movable along the valve axis. The poppet 40 has a sealing face 42, which in the principal embodiment is conical and which seats on the sealing edge 32 to close the tyre valve 1. From the closed position the poppet 40 travels away from the sealing edge 32 through a normal stroke 114 and, beyond the normal stroke 114, through an over-travel 116.

[0119] The poppet 40 has a purge shoulder 44. The purge shoulder 44 is a surface of the poppet 40 that is separate from the sealing face 42. In the principal embodiment the purge shoulder 44 is located below the sealing face 42, that is, it is spaced from the sealing face 42 along the valve axis. In some embodiments the purge shoulder 44 is annular. The purge shoulder 44 is spaced from the seat ring 30 when the poppet 40 is at rest and throughout the normal stroke 114. The purge shoulder 44 contacts the seat ring 30 only during the over-travel 116, in which it applies an axial load to the seat ring 30.

[0120] Because the purge shoulder 44 is spaced from the seat ring 30 up to the end of the normal stroke 114, the seat ring 30 is loaded by the purge shoulder 44 only in the over-travel 116. At rest the seat ring 30 carries on its sealing edge 32 the load with which the sealing face 42 is held closed. That load is distinct from the axial load applied by the purge shoulder 44.

[0121] The return spring 50 biases the poppet 40 towards the closed position. In the principal embodiment the return spring 50 is a compression spring arranged under the poppet 40. The return spring 50 has a two-rate characteristic: a first spring stage 52 of lower rate acts through the normal stroke 114, and a second spring stage 54 of higher rate must be overcome for the poppet 40 to reach the over-travel 116. The return spring 50 thereby sets two values of the pressure differential. The first is a crack pressure 110, at which the poppet 40 first lifts off the sealing edge 32 against the first spring stage 52. The second is an over-travel differential 112, greater than the crack pressure 110, which overcomes the second spring stage 54 and produces the over-travel 116.

[0122] The two-rate characteristic separates the over-travel differential 112 from the crack pressure 110. Inflation at a pressure differential between the two takes place through the normal stroke 114 alone, and the over-travel 116 is reached only at the higher pressure differential.

[0123] FIG. 5 shows the two-rate characteristic qualitatively. The horizontal axis represents the travel of the poppet 40 from the closed position and the vertical axis represents the force on the poppet 40. The first spring stage 52 appears as a segment of lower slope extending over the normal stroke 114, and the second spring stage 54 as a segment of higher slope extending over the over-travel 116. The crack pressure 110 and the over-travel differential 112 are marked on the curve. FIG. 5 is schematic: it carries no scale and represents no measured values. FIG. 5 does not depend on how the two-rate characteristic is produced, which may be by a single spring or by more than one spring or biasing element, as described under "Return bias and load-applying surface".

[0124] The normal stroke 114 and the over-travel 116 are two successive ranges of the travel of the same poppet 40. In the principal embodiment the end of the normal stroke 114 can be identified in the valve core 5 in two ways. It is the travel at which the first spring stage 52 gives way to the second spring stage 54, which appears in FIG. 5 as the change of slope. It is also the limit up to which the purge shoulder 44 remains spaced from the seat ring 30; beyond that limit, in the over-travel 116, the purge shoulder 44 bears on the seat ring 30.

[0125] A main passage 26 is the path by which inflation air flows into the tyre chamber 80 when the poppet 40 is lifted off the sealing edge 32. The main passage 26 runs past the sealing edge 32, between the sealing edge 32

and the sealing face 42. In FIG. 1 the poppet 40 is seated, so that the main passage 26 is closed and is indicated as a path.

[0126] A purge gap 60 is a secondary flow path into the tyre chamber 80 that is separate from the main passage 26. The purge gap 60 is bounded by the outer surface of the seat ring 30 and by a bore surrounding the seat ring 30. The purge gap 60 is closed at rest and is open when the seat ring 30 has undergone the radial expansion. In the principal embodiment the bore surrounding the seat ring 30 is the core body bore 22, and the purge gap 60 is an annular clearance between the outer surface of the seat ring 30 and the core body bore 22. In FIG. 1 the purge gap 60 is closed and is drawn as a single line.

[0127] In some embodiments the bore surrounding the seat ring 30 is the bore of the valve stem 10, so that the purge gap 60 lies between the seat ring 30 and the bore of the valve stem 10. This arrangement is described further under "Purge gap arrangements and retention of the seat ring".

[0128] The purge gap 60 differs from the main passage 26 in where it runs and in which surfaces it sweeps. The main passage 26 runs between the poppet 40 and the seat ring 30, past the sealing edge 32. The purge gap 60 runs along the outer surface of the seat ring 30, between the seat ring 30 and the bore surrounding it. Air flowing through the purge gap 60 passes over the outer surface of the seat ring 30 and over the surrounding bore. The poppet 40 is not one of the boundaries of the purge gap 60. A main passage 26 that is made larger, whether by further travel of the poppet 40 or by deflection of the seat ring 30, remains the main passage 26 and is not a purge gap 60.

[0129] Arrangements for retaining the seat ring 30 in the core body 20 while leaving the seat ring 30 free to undergo the radial expansion, and the exit of the air purge into the tyre chamber 80, are described under "Purge gap arrangements and retention of the seat ring".

[0130] In some embodiments a lock nut 70 is threaded onto the stem of the poppet 40 outside the core body 20. The lock nut 70 is conventional. The lock nut 70 is used to close the poppet 40 manually for storage and transport, and it plays no part in the opening of the purge gap 60.

Principal embodiment: operation

[0131] The operation of the principal embodiment over one inflation cycle is described with reference to FIGS. 1, 2 and 5. An inflation cycle is one application and removal of the pressure differential by the pump head 90. The steps of use are set out under "Use: cleaning during an inflation cycle".

[0132] At rest, as shown in FIG. 1, the return spring 50 holds the sealing face 42 of the poppet 40 on the sealing edge 32 of the seat ring 30, and the main passage 26 is closed. The purge shoulder 44 is spaced from the seat ring 30. The seat ring 30 is in its at-rest shape and the purge gap 60 is closed, so that no air purge passes through the purge gap 60 and the tyre valve 1 retains the air in the tyre chamber 80.

[0133] When the pump head 90 is coupled to the tyre valve 1 and the pressure differential exceeds the crack pressure 110, the poppet 40 lifts off the sealing edge 32 against the first spring stage 52 and travels through the normal stroke 114. Air flows through the main passage 26 into the tyre chamber 80.

[0134] Throughout the normal stroke 114 the purge shoulder 44 remains spaced from the seat ring 30. The seat ring 30 is not loaded by the purge shoulder 44 and keeps its at-rest shape, and the purge gap 60 remains closed. The geometry of the sealing edge 32 is therefore the same after a normal stroke 114 as before it. At a pressure differential below the over-travel differential 112 the tyre chamber 80 is inflated through the main passage 26 alone.

[0135] When the pressure differential exceeds the over-travel differential 112, the poppet 40 overcomes the second spring stage 54 and travels beyond the normal stroke 114 into the over-travel 116, as shown in FIG. 2. In the over-travel 116 the purge shoulder 44 bears on the seat ring 30 and applies an axial load to it. Under that axial load the seat ring 30 undergoes radial expansion, and the purge gap 60 opens.

[0136] The convoluted wall 34 converts the axial load into the radial expansion. In some embodiments the convolution 36 flattens under the axial load, and FIG. 2 shows the convolution 36 flattened in comparison with

FIG. 1. The convoluted wall 34 then deforms by bending at the convolution 36. The radial expansion is obtained from this change in the shape of the convoluted wall 34 and not from a uniform circumferential stretching of the convoluted wall 34.

[0137] The purge gap 60 is opened by the mechanical load that the poppet 40 applies through the purge shoulder 44. The purge gap 60 is not opened by the flow of air acting on the seat ring 30. Whether the poppet 40 reaches the over-travel 116 depends on the pressure differential applied by the pump head 90. While the poppet 40 is in the over-travel 116, the purge shoulder 44 holds the seat ring 30 in its expanded shape whatever the rate at which air passes through the purge gap 60.

[0138] With the purge gap 60 open, air at the pressure delivered by the pump head 90 flows through the purge gap 60 into the tyre chamber 80. This flow is the air purge. The main passage 26 remains open during the over-travel 116, so that air flows into the tyre chamber 80 through the main passage 26 and through the purge gap 60 at the same time, as indicated by the two flow arrows in FIG. 2. The air purge passes over the outer surface of the seat ring 30 and over the bore surrounding it, and acts on sealant film present on those surfaces so as to carry it into the tyre chamber 80.

[0139] When the pump head 90 is removed, the pressure differential collapses and the return spring 50 returns the poppet 40. As the poppet 40 leaves the over-travel 116 the purge shoulder 44 lifts from the seat ring 30 and the axial load is removed. The seat ring 30 undergoes shape recovery, returning to its at-rest shape, and the purge gap 60 closes. The poppet 40 then returns through the normal stroke 114 and the sealing face 42 seats again on the sealing edge 32.

[0140] In some embodiments, in particular those in which the seat ring 30 is of a superelastic material, the shape recovery gives a second cleaning action, the snap-back wipe. The shape recovery of a superelastic seat ring 30 is fast and stiff. It is believed that on shape recovery the seat ring 30 moves across the sealing face 42 of the poppet 40 and shears partially cured sealant film from it. The surfaces concerned in the snap-back wipe are the seat ring 30 and the sealing face 42.

[0141] An inflation cycle in which the poppet 40 reaches the over-travel 116 therefore provides the air purge while the purge gap 60 is open and, in the embodiments that have it, the snap-back wipe when the pump head 90 is removed. Both cleaning actions take place in the course of the inflation cycle, without a tool and without removal of the valve core 5 from the valve stem 10. The air purge takes place before the shape recovery and does not require the snap-back wipe. The snap-back wipe is an additional cleaning action that is present in some embodiments.

[0142] The test reported in Example 1 was carried out on valve cores according to the principal embodiment having a seat ring 30 of superelastic NiTi alloy. That test does not isolate the air purge from the snap-back wipe, and no statement is made as to the share of each cleaning action in the result reported there. When those valve cores were inspected at the end of that test, the sealing edge 32 and the sealing face 42 were clean, and cured sealant film was present on the outer surface of the seat ring 30 in the region of the purge gap 60.

[0143] In the test reported in Example 1, a snap was heard on removal of the pump head 90 after each inflation of the valve cores according to the principal embodiment. The snap was taken in that test as an indication that the poppet 40 had reached the over-travel 116. The snap is recorded as an observation, and no embodiment requires it.

Return bias and load-applying surface

[0144] The return spring 50 is an example of a return bias, that is, of an arrangement that biases the poppet 40 towards the closed position. A return bias having a two-rate characteristic can be constructed in several ways, which are described in the following paragraphs. Each construction provides a first spring stage 52 that the crack pressure 110 overcomes and a second spring stage 54 that the over-travel differential 112 overcomes. The rates of the first spring stage 52 and of the second spring stage 54 are selected so that the crack pressure 110 and the over-travel differential 112 have the values described under "Parameters and ranges".

[0145] In some embodiments the return spring 50 is a single compression spring of variable pitch. The closely pitched coils of the variable-pitch spring close solid during the normal stroke 114. The widely pitched coils that

remain active are fewer and so give a higher rate, and they form the second spring stage 54. The first spring stage 52 and the second spring stage 54 are thereby obtained from one part, and the parts that adjoin the return spring 50 are unchanged. FIGS. 1 and 2 show a return spring 50 of this construction by way of example.

[0146] In some embodiments the return bias comprises a soft spring and a stiff spring arranged in series, with a stop that takes the soft spring out of the load path at the end of the normal stroke 114. The soft spring provides the first spring stage 52 and sets the crack pressure 110. The stiff spring provides the second spring stage 54 and sets the over-travel differential 112.

[0147] In some embodiments the return bias comprises nested springs that act in parallel with staggered engagement. An outer spring acts over the whole travel of the poppet 40. A shorter inner spring engages only at the end of the normal stroke 114, so that the rate in the over-travel 116 is the sum of the rates of the two springs. This construction occupies less axial length than two springs in series.

[0148] In some embodiments the return bias comprises a return spring 50 of a single rate and an elastomeric bumper that the poppet 40 reaches at the end of the normal stroke 114. The elastomeric bumper provides the second spring stage 54. The material of the elastomeric bumper is selected to withstand the sealant, including sealant that contains ammonia, and to resist compression set, since compression set would shift the over-travel differential 112.

[0149] In some embodiments the return bias comprises a soft coil spring stacked with a disc spring, the disc spring being engaged at the end of the normal stroke 114. The disc spring provides a second spring stage 54 that is stiff and occupies little axial length.

[0150] In some embodiments a detent holds the poppet 40 at the end of the normal stroke 114 until the over-travel differential 112 overcomes the detent. The detent is, for example, a spring-loaded ball, or a magnet and a keeper. In these embodiments the crack pressure 110 and the over-travel differential 112 are separated by a step in force in place of a change of rate, and the entry of the poppet 40 into the over-travel 116 is abrupt.

[0151] Where the two-rate characteristic is provided by a single spring, the over-travel differential 112 does not depend on the seating of two springs relative to one another. In some embodiments a single variable-pitch spring is used for that reason in preference to a construction having two springs.

[0152] In some embodiments the return spring 50 is subjected to a pre-setting step, that is, a stabilising step, before it is assembled into the valve core 5. The pre-setting step is directed to limiting relaxation of the return spring 50 under repeated compression, since relaxation would cause the crack pressure 110 and the over-travel differential 112 to drift in service.

[0153] In some embodiments the normal stroke 114 ends at a spring-loaded stop. The poppet 40 meets a stop washer at the end of the normal stroke 114, and the stop washer is itself held by a spring that is stiffer than the return spring 50. The over-travel 116 is the travel in which the poppet 40 pushes the stop washer back against the stiffer spring. The stop washer defines the end of the normal stroke 114 and the stiffer spring sets the over-travel differential 112, and the return spring 50 can then have a single rate.

[0154] The purge shoulder 44 is an example of a load-applying surface, that is, of a surface that is moved by the poppet 40 in the over-travel 116 and that applies the axial load to the seat ring 30 or to another radially deflectable element. In the principal embodiment the load-applying surface is the purge shoulder 44 formed on the poppet 40 itself.

[0155] In some embodiments the load-applying surface is provided on a collar that is a separate part carried by the poppet 40. The collar is pressed, crimped or threaded onto the poppet 40 below the sealing face 42. The position of the collar on the poppet 40 sets the over-travel distance, so that the over-travel distance can be set when the valve core 5 is assembled.

[0156] In some embodiments the purge shoulder 44 is conical in place of a flat annulus. A conical purge shoulder 44 is used with forms of the radially deflectable element that are spread by a cone, namely the slotted sleeve, the split seat ring and the tapered seat ring described under "Forms of the radially deflectable element".

[0157] In some embodiments the axial load is applied to the seat ring 30 through the return bias. A spring seat washer is arranged between the return spring 50 and the poppet 40 and bears on the seat ring 30 once the first spring stage 52 has been taken up. In these embodiments the load-applying surface is a surface of the spring seat washer, and the spring seat washer, not the poppet 40, touches the seat ring 30.

[0158] In each of these embodiments the load-applying surface is moved by the poppet 40, and the radially deflectable element is loaded by the load-applying surface only in the over-travel 116.

Tyre valve patterns and product forms

[0159] In the principal embodiment the tyre valve 1 is a Presta-pattern tyre valve. The valve stem 10 is a Presta-pattern valve stem of brass with an outer diameter of 6 mm and a threaded bore that receives the valve core 5. The valve stem 10 is conventional. The Presta pattern is an example of a pattern of the tyre valve 1.

[0160] In the principal embodiment the valve core 5 is a removable threaded valve core. The core body 20 is screwed into the threaded bore of the valve stem 10, and the valve core 5 can be unscrewed from the valve stem 10 and replaced as a unit.

[0161] In some embodiments the valve core is integral with the valve stem, as shown in FIG. 13B, in which the valve core integral with the valve stem is designated 230. The core body is formed in the valve stem 10 and is not threaded into it. The seat ring 30, the poppet 40 with its purge shoulder 44, and the return spring 50 are as in the principal embodiment. A valve core of this form is not removed from the valve stem 10 for service.

[0162] The tyre chamber 80 is the pressurised volume into which the tyre valve 1 admits air. In some embodiments the tyre chamber 80 is a bicycle inner tube. In some embodiments the tyre chamber 80 is the cavity formed by a tubeless tyre and a rim. In each case the tyre chamber 80 contains sealant in use, and the sealant is introduced into the tyre chamber 80 through the tyre valve 1.

[0163] In some embodiments the valve core 5 is supplied as a unit separate from the valve stem 10 and from the tyre chamber 80, for fitting to an existing valve stem 10 in place of a valve core that has been removed. A valve core 5 so supplied comprises the core body 20, the seat ring 30, the poppet 40 and the return spring 50. Where the bore surrounding the seat ring 30 is the core body bore 22, a valve core 5 so supplied contains both boundaries of the purge gap 60.

[0164] In some embodiments the seat ring 30 is supplied separately as a replacement part for fitting into a core body 20.

Forms of the radially deflectable element

[0165] The seat ring 30 described under "Principal embodiment: structure" is one form of a radially deflectable element, as that term is defined under "Definitions and interpretation". The radially deflectable element is also embodied by the polymer seat ring 130 shown in FIG. 8 and by the wiper ring 144 of the two-piece seat shown in FIG. 9A and FIG. 9B, and in some embodiments by the further forms shown in FIG. 11A to FIG. 11C. In each form, a load-applying surface moved by the poppet 40, for example the purge shoulder 44, bears on the radially deflectable element only in over-travel, the axial load so applied deflects the element radially, and the element returns by shape recovery when the load is removed.

[0166] In some embodiments the radially deflectable element carries the sealing edge, as the seat ring 30 carries the sealing edge 32 and the polymer seat ring 130 carries the sealing edge 132. In other embodiments the radially deflectable element is separate from the element that carries the sealing edge, as in the two-piece seat, in which the rigid seat 140 carries the sealing edge 142 and the wiper ring 144 is the radially deflectable element.

[0167] In some embodiments the radially deflectable element has a load-converting geometry, that is, a geometry by which the axial load applied in over-travel is converted into radial expansion. The convoluted wall 34 of the seat ring 30 is the load-converting geometry of the principal embodiment. In some embodiments the conversion is made by the shape of the radially deflectable element, by the shape of the load-applying surface that bears on it, or by both.

[0168] The effect of the load-converting geometry is the opening of a flow path. The axial load is converted into radial expansion, and the radial expansion opens the purge gap 60, which is a secondary flow path distinct from the main passage 26 and closed at rest. A wavy or undulating shape of a resilient metal seal is described in US11549591B1, where the shape pre-loads a seat interface to provide a seating and sealing force. The load-converting geometry described here has a different effect: it acts when the poppet 40 is in over-travel, and its deformation opens the purge gap 60 through which the air purge passes.

[0169] In some embodiments the convoluted wall 34 has one convolution 36, as shown in FIG. 4A. The convolution 36 is a single shallow wave in the section of the convoluted wall 34, between the sealing edge 32 and the end of the seat ring 30 that is seated against the locating shoulder 24. Axial load applied to the seat ring 30 in over-travel is translated by the convolution 36 into radial expansion of the convoluted wall 34.

[0170] In other embodiments the convoluted wall 34 has two convolutions, as shown in FIG. 4B: a convolution 36 nearer the sealing edge 32 and a convolution 38 nearer the fit band 35. The two convolutions lie in series along the convoluted wall 34 and together translate the axial load into radial expansion.

[0171] In FIG. 4A and FIG. 4B the wall thickness is marked t, the free length is marked L and the convolution amplitude is marked a. Values of these dimensions are given under "Parameters and ranges".

[0172] In some embodiments the convolution 36 is formed in the section of the wall by shape-setting. A plain tube wall is pressed into one or two circumferential waves in a shape-setting fixture, no pattern is cut in the wall, and laser cutting serves only to trim the ends of the seat ring 30. Under axial load the waves flatten and the wall moves outward.

[0173] In some embodiments the load-converting geometry is a strut pattern cut in the wall by laser cutting, for example a zig-zag pattern or a pattern of diamond-shaped cells. Under axial load the strut pattern shortens axially and bulges radially, so that the radial expansion is obtained by bending of the struts. In such embodiments the cells of the strut pattern are closed at rest, and the cut portion of the wall provides a route for the purge flow when the wall deforms, as described under "Purge gap arrangements and retention of the seat ring".

[0174] In some embodiments the load-converting geometry is a pre-bowed wall, that is, a wall having a single outward bow along the free length. The bow increases under axial load in the manner of a pre-buckled column. The pre-bowed wall corresponds to the convoluted wall 34 with one convolution 36, written as a bow, and its purpose is a radial expansion that increases progressively with the axial load.

[0175] In some embodiments the load-converting geometry is a helical slot through the wall along part of the free length. Axial compression unwinds the helix slightly and increases the diameter of the wall, and the helical slot gives the wall an axial compliance. The helical slot is itself a route for the purge flow when it is open.

[0176] In some embodiments the convolution 36 has a root radius sized to reduce the concentration of strain at the root of the convolution 36, where repeated radial expansion and shape recovery concentrate the strain.

[0177] In some embodiments the convolution amplitude is chosen towards the lower end of the range given under "Parameters and ranges", to reduce the peak strain in the convoluted wall 34. No service life is stated for any form of the radially deflectable element; the extent of the test carried out is stated under "Examples and data".

[0178] In some embodiments the radially deflectable element is a slotted sleeve 170, shown in FIG. 11A. The slotted sleeve 170 is a thin-walled cylinder in which axial slots running from one end form fingers 172. The purge shoulder 44 is conical in these embodiments and, in over-travel, enters between the fingers 172 and spreads them outward towards the core body bore 22. The radial expansion is obtained by bending of the fingers 172 and not by hoop strain of a continuous wall. The axial slots between the fingers 172 are a route for the purge flow when the fingers 172 are spread.

[0179] In some embodiments the radially deflectable element is a split seat ring 180, shown in FIG. 11B. The split seat ring 180 has a C-shaped section with one axial split 182, and opens circumferentially when a cone on the poppet 40 enters it in over-travel. The purpose of the axial split 182 is a change of diameter obtained with little strain of the material. In such embodiments the axial split 182 is closed at rest, or the seal at rest is made on a face beside the axial split 182, so that the axial split 182 does not interrupt the seal at rest.

[0180] In some embodiments the radially deflectable element is a tapered seat ring 190, shown in FIG. 11C. The tapered seat ring 190 has a conical inner surface, the taper 192, and the poppet 40 has a matching taper 194 in place of the purge shoulder 44. In over-travel the matching taper 194 wedges the tapered seat ring 190 outward, so that the axial load is converted into radial expansion without a convolution. In some such embodiments the angle of the taper 192 is chosen greater than the friction angle between the taper 192 and the matching taper 194, so that the tapered seat ring 190 is released when the poppet 40 returns.

[0181] In some embodiments the element deflected by the load-applying surface is a disc spring of conical form, the rim of which seals on a face at rest. In over-travel the load-applying surface flattens the disc spring, the rim lifts from the face, and a purge gap opens at the face. In these embodiments the deflection is axial and the purge gap is a gap at a face, so that the element is referred to as a deflectable element and its deflection is not a radial expansion. The disc spring also provides a second spring stage. No figure shows this form and no test of it is reported.

[0182] In some embodiments the deformation of the deflectable element that opens the purge gap is radial, axial, tilting or a combination of these. Radial expansion is the deformation of the principal embodiment.

[0183] In some embodiments the valve core 5 has a two-piece seat, shown at rest in FIG. 9A and in over-travel in FIG. 9B. The two-piece seat comprises a rigid seat 140 and a wiper ring 144. The rigid seat 140 carries the sealing edge 142, on which the sealing face 42 of the poppet 40 seats. The wiper ring 144 is a separate thin ring located below the rigid seat 140. In some embodiments the rigid seat 140 is made of brass and the wiper ring 144 is made of stainless spring steel.

[0184] At rest the sealing face 42 is seated on the sealing edge 142 of the rigid seat 140, so that the seal at rest is made on the rigid seat 140 and does not depend on the wiper ring 144. In over-travel the purge shoulder 44 bears on the wiper ring 144 and deflects it outward, as shown in FIG. 9B. When the load is removed the wiper ring 144 returns by shape recovery and wipes on its return.

[0185] The two-piece seat separates the sealing function, which is on the rigid seat 140, from the function of the element deflected by the purge shoulder 44, which is on the wiper ring 144. The two-piece seat has one more part than the seat ring 30 and a greater axial length, and may be simpler to make. No two-piece seat has been built or tested.

[0186] Of the forms described in this section, the seat ring 30 having the convoluted wall 34 and made of superelastic NiTi alloy has been built and tested, as reported under "Examples and data". The other forms are described as embodiments, and no test result is stated for them.

Purge gap arrangements and retention of the seat ring

[0187] The purge gap 60 is bounded by an outer surface of the seat ring 30 and by a bore surrounding the seat ring 30, as described under "Principal embodiment: structure". This section describes embodiments in which the seat ring 30 is retained in the core body 20 while a portion of it remains free to deflect, and embodiments of the route taken by the purge flow.

[0188] In some embodiments the interference fit acts on a fit band 35 of the seat ring 30 and not on the whole of its outer surface. The fit band 35 is a short band of the outer surface next to the locating shoulder 24, and the remaining length of the outer surface is a free portion 37, as shown in FIG. 4A and FIG. 4B. The core body bore 22 has a core body bore relief 202, for example a counterbore or an undercut, which begins just past the fit band 35 and extends along the free portion 37, as shown in FIG. 12A.

[0189] In these embodiments the seat ring 30 closes the purge gap 60 at rest by line contact at one end of the core body bore relief 202. In over-travel the free portion 37 expands into the core body bore relief 202 and the purge gap 60 opens, while the fit band 35 remains retained in the core body bore 22.

[0190] The effect of the fit band 35 and the free portion 37 is that the retention of the seat ring 30 and the radial expansion are carried by different portions of the same outer surface. The core body 20 has one more machined diameter than a core body having a plain core body bore 22.

[0191] In some embodiments the purge gap 60 is bounded by the outer surface of the seat ring 30 and by the bore of the valve stem 10. In one such arrangement the seat ring 30 sits at the end of the core body 20, so that the outer surface of the seat ring 30 faces the bore of the valve stem 10. The valve stem 10 then provides one boundary of the purge gap 60, and that boundary remains in the valve stem 10 when the valve core 5 is removed.

[0192] In some embodiments the purge flow passes through the wall of the seat ring 30. In these embodiments wall slots 204, or perforations, are provided through the convoluted wall 34 in the free portion 37, below the sealing edge 32, as shown in FIG. 12B. The wall slots 204 are closed at rest, either because they lie against the bore surrounding the seat ring 30 or because their edges abut, and they open when the convoluted wall 34 deforms in over-travel.

[0193] In the embodiments having the wall slots 204, the purge flow sweeps the inner surface and the outer surface of the seat ring 30, and the outer surface of the seat ring 30 can be retained along its whole length. The purge gap is then not annular.

[0194] In some embodiments the purge gap is a secondary flow path opened by deformation of the seat ring 30 and located around the seat ring 30, through it or within it. The secondary flow path is continuous in some embodiments, as in the annular purge gap 60 around the seat ring 30, and segmented in others, for example a purge gap formed at one or more lobes of the seat ring 30 or at several wall slots 204 in the convoluted wall 34.

[0195] In the embodiments having the slotted sleeve 170, the axial slots between the fingers 172 provide the route for the purge flow. In the embodiments having the strut pattern or the helical slot, the cut portion of the wall provides it. These forms are described under "Forms of the radially deflectable element".

[0196] The purge flow leaves the purge gap 60 into the tyre chamber 80. Air at pump pressure flows through the open purge gap 60, across the surfaces on which the sealant film accumulates, and into the tyre chamber 80, alongside the flow through the main passage 26, as shown by the flow arrows of FIG. 2. This flow is the air purge.

[0197] In the principal embodiment the seat ring 30 is pressed into the core body 20 with a light interference fit and is seated against the locating shoulder 24. The amount of the interference fit is given under "Manufacture and assembly".

[0198] In some embodiments the seat ring 30 is retained by axial clamping in place of the interference fit. The seat ring 30 is clamped between the locating shoulder 24 and a retainer sleeve, which is pressed or threaded into the core body 20. The seat ring 30 is then retained without radial preload on the free portion 37.

[0199] In some embodiments the seat ring 30 is held against the locating shoulder 24 by staking, the rim of the core body 20 being staked over the end of the seat ring 30. This retention likewise applies no radial preload to the free portion 37, and it adds one operation where the retainer sleeve adds one part.

[0200] In some embodiments the valve core 5 has a positive stop which fixes the position of the sealing face 42 against the sealing edge 32 at rest, independently of the shape recovery of the seat ring 30. The seal at rest then does not depend solely on complete shape recovery of the seat ring 30 after repeated over-travel.

[0201] In some embodiments an anti-rotation key or a flat is provided between the seat ring 30 and the core body bore 22, in addition to the interference fit, to hold the seat ring 30 against axial shift and against rotation in the core body bore 22.

[0202] In some embodiments an alignment chamfer is provided on the locating shoulder 24 or on the core body bore 22. The alignment chamfer aligns the seat ring 30 during press-fitting, so that the seat ring 30 is not seated at an angle to the locating shoulder 24, and the contact at the sealing edge 32 and the radial expansion are uniform around the seat ring 30.

[0203] As reported under "Examples and data", a cured sealant film was found on the outer surface of the seat ring 30 in the region of the purge gap 60 at the end of the test. In some embodiments electropolishing is applied to the surfaces facing the purge gap 60, namely the outer surface of the seat ring 30 and, where needed, the core body bore 22, to reduce adhesion of the sealant in that region.

[0204] The core body bore relief 202, the wall slots 204, the retention by axial clamping or by staking, the positive stop, the anti-rotation key and the alignment chamfer are described as embodiments. They are not stated to be the construction of the valve cores tested under "Examples and data", and no test result is stated for them.

Materials and shape recovery

[0205] In the principal embodiment the seat ring 30 is made of superelastic NiTi alloy of nominally 55.8 wt% Ni, in the form of drawn tube which is shaped by shape-setting and finished by electropolishing. The austenite finish temperature of the NiTi alloy is below 5 °C, so that the seat ring 30 is superelastic at riding temperatures.

[0206] In some embodiments the radially deflectable element is made of an elastically recovering material having radial compliance. Superelastic NiTi alloy, a polymer and a spring-tempered metal are examples of such a material. Shape recovery of the radially deflectable element does not require a transformation of the material.

[0207] In some embodiments the seat ring 30 is made of a ternary superelastic alloy, for example a Ni–Ti–Cr alloy or a Ni–Ti–Cu alloy, in place of the binary NiTi alloy. The addition of chromium shifts the transformation temperature, and the addition of copper narrows the hysteresis. The parameters of shape-setting are adapted to the alloy.

[0208] In some embodiments the seat ring 30 is made of a copper-based alloy that is superelastic, for example a Cu–Al–Ni alloy or a Cu–Zn–Al alloy. The recoverable strain of such alloys is lower than that of NiTi alloy, and their compatibility with an ammonia-bearing sealant is not established.

[0209] In some embodiments the valve core 5 has a polymer seat ring 130 in place of the seat ring 30, as shown in FIG. 8. The polymer seat ring 130 has a sealing edge 132 and a convoluted wall 134, the convoluted wall 134 being the same as the convoluted wall 34 of the seat ring 30. In some embodiments the polymer seat ring 130 is made of polyether ether ketone (PEEK). The core body 20, the core body bore 22, the locating shoulder 24, the poppet 40 with the sealing face 42 and the purge shoulder 44, the return spring 50 and the purge gap 60 are as in the principal embodiment.

[0210] The polymer seat ring 130 has some radial compliance and a lower cost than the seat ring 30 of superelastic NiTi alloy. Its shape recovery is slower and the force of its return is lower, so that the snap-back wipe is weaker. No polymer seat ring 130 has been built or tested.

[0211] In some embodiments the polymer seat ring 130 is made of another high-modulus thermoplastic, for example polyphenylene sulphide (PPS) or polyetherimide (PEI), in place of PEEK. The chemical compatibility of each such polymer with an ammonia-bearing sealant is to be confirmed for that polymer.

[0212] In some embodiments the radially deflectable element is made of a spring-tempered metal, for example spring-tempered stainless steel or copper–beryllium, and has a form in which the deflection is dominated by bending, such as the slotted sleeve 170 or the strut pattern. The bending-dominated deflection keeps the material within its elastic limit, so that the shape recovery is the elastic recovery of a metal that is not superelastic. A spring-tempered metal is not proposed for a continuous convoluted wall 34.

[0213] In some embodiments the shape recovery is assisted by a garter spring, that is, a circumferential coil spring placed around the slotted sleeve 170, the split seat ring 180 or the polymer seat ring 130. The garter spring supplies a closing force, so that the closing of the purge gap does not depend only on the stiffness of the radially deflectable element. The garter spring lies in the region of the purge gap and passes the purge flow between its coils.

[0214] In some embodiments the austenite finish temperature of the NiTi alloy is selected with a margin below the coldest temperature expected at the tyre valve 1 in use, for example the coldest riding or storage climate anticipated, or the temperature reached at the tyre valve 1 when a CO■ cartridge is discharged through it. The purpose of the margin is that the seat ring 30 remains superelastic at that temperature. No value of the margin is stated, and no test at low temperature is reported.

[0215] In the principal embodiment the core body 20 and the valve stem 10 are made of free-machining brass, for example of grade CW614N. In some embodiments the core body 20 is made of stainless steel, for example of

303 grade, in place of brass, for corrosion resistance where the sealant is an ammonia-bearing sealant. A core body 20 of stainless steel has been machined; no test of it with sealant is reported.

[0216] In the principal embodiment the poppet 40 is made of brass. In some embodiments the sealing face 42 of the poppet 40 is of hard-anodised aluminium, to resist wear and scoring of the sealing face 42 by repeated seating and to resist corrosion. No poppet 40 having such a sealing face 42 has been built.

[0217] In the principal embodiment the return spring 50 is made of stainless music wire. The effect described does not depend on the material of the return spring 50.

[0218] In some embodiments the seat ring is a thermal variant 150, shown schematically in FIG. 10, made of a NiTi alloy trained with two-way shape memory and having an austenite finish temperature of approximately 35 °C to 40 °C. Warming of the thermal variant 150, for example by hand, opens the purge gap 60 without a pump, and FIG. 10 accordingly shows the poppet 40 with the sealing face 42 seated and the purge gap 60 open. The thermal variant 150 has not been built and no data exist for it.

Combinations

[0219] The features of the embodiments described above may be combined as set out in this section. The combinations are stated for the seat ring 30 in general, for the seat ring 30 having the convoluted wall 34, for the seat ring 30 together with the purge shoulder 44 and the return spring 50 having the two-rate characteristic, for the seat ring 30 made of superelastic NiTi alloy, for the polymer seat ring 130 and for the two-piece seat.

[0220] In some embodiments the core body 20 is formed from stainless steel, for example of 303 grade, in place of brass, for corrosion resistance where the sealant is ammonia-bearing. This variant may be combined with each of the forms listed in the preceding paragraph, including a seat ring 30 made of superelastic NiTi alloy, a polymer seat ring 130 and a two-piece seat.

[0221] In some embodiments a lock nut 70 is provided on the poppet 40 to close the poppet 40 manually for storage or transport. The lock nut 70 may be provided in combination with any of the forms listed above, including a seat ring 30 made of superelastic NiTi alloy, a polymer seat ring 130, and a two-piece seat having a rigid seat 140 and a wiper ring 144.

[0222] In some embodiments the convoluted wall 34 has one convolution 36, and in other embodiments it has two convolutions. Either number of convolutions may be combined with a seat ring 30 made of superelastic NiTi alloy or with a polymer seat ring 130.

[0223] In some embodiments the sealing face 42 of the poppet 40 is of hard-anodised aluminium. This variant may be combined with a core body 20 of brass or of stainless steel, and with a seat ring 30 made of superelastic NiTi alloy.

[0224] In some embodiments the seat ring 30 is finished by electropolishing after shape-setting and quenching, to remove the oxide layer and the heat-affected edge. The electropolishing is described for a seat ring 30 formed from a metal such as superelastic NiTi alloy.

[0225] The austenite finish temperature below 5 °C applies where the seat ring 30 is made of a NiTi alloy. The thermal variant 150 likewise applies where the seat ring is made of a NiTi alloy, and requires in addition that the NiTi alloy be trained with two-way shape memory.

[0226] In one combination the valve core 5 has the seat ring 30 made of superelastic NiTi alloy and finished by electropolishing, and the core body 20 made of stainless steel. The seat ring 30 of this combination is that of the valve cores tested under "Examples and data", and the core body 20 of stainless steel is provided for corrosion resistance with an ammonia-bearing sealant.

[0227] In another combination the valve core 5 has the seat ring 30 made of superelastic NiTi alloy and the lock nut 70. This is the form in which the valve core 5 can be closed manually for storage and transport.

[0228] In another combination the valve core 5 has the polymer seat ring 130 and the core body 20 made of stainless steel. In this combination the material of the polymer seat ring 130 and the material of the core body 20

both differ from those of the principal embodiment, the polymer seat ring 130 giving the lower cost and the core body 20 of stainless steel giving the corrosion resistance.

[0229] In another combination the valve core 5 has the seat ring 30 made of superelastic NiTi alloy, the convoluted wall 34 having one convolution 36 or two convolutions, and the sealing face 42 of hard-anodised aluminium. This combination joins the form of the convoluted wall 34 with a variant of the material of the poppet 40.

[0230] In another combination the valve core 5 has the two-piece seat and the core body 20 made of stainless steel. The material of the core body 20 does not depend on whether the sealing edge is carried by the seat ring 30 or by the rigid seat 140.

[0231] Further combinations are stated where the features concerned are described. The purge shoulder 44 of conical form is combined with the slotted sleeve 170, the split seat ring 180 or the tapered seat ring 190. The spring-tempered metal is combined with the slotted sleeve 170 or with the strut pattern. The garter spring is combined with the slotted sleeve 170, the split seat ring 180 or the polymer seat ring 130. The fit band 35 and the free portion 37 are combined with the core body bore relief 202, and the retention by axial clamping or by staking is combined with the free portion 37. The wall slots 204 are combined with retention of the outer surface of the seat ring 30 along its whole length.

Parameters and ranges

[0232] The dimensions, travels and pressures of the valve core 5 described under "Principal embodiment: structure" are set out in this section. For each parameter one value has been built and tested, namely the value used in the prototype valve cores of Example 1. Each range stated together with such a value is the range within which the inventors consider that the valve core 5 operates as described. The ranges have not been explored by test; only the prototype value of each parameter has been tested.

[0233] In some embodiments the seat ring inner diameter, which is the internal diameter of the seat ring 30, is from 2.0 mm to 2.6 mm. In the prototype valve cores the seat ring inner diameter was 2.3 mm, which is the internal diameter of a tube of NiTi alloy having an outside diameter of 2.7 mm and a wall of 0.20 mm.

[0234] In some embodiments the wall thickness, which is the thickness of the convoluted wall 34, is from 0.15 mm to 0.30 mm. In the prototype valve cores the wall thickness was 0.20 mm.

[0235] In some embodiments the free length, which is the axial length of the seat ring 30 when unloaded, is from 2.5 mm to 4.0 mm. In the prototype valve cores the free length was 3.0 mm.

[0236] In some embodiments the convolution amplitude, which is the height of a convolution 36 measured from the mean line of the convoluted wall 34, is from 0.08 mm to 0.20 mm. In the prototype valve cores the convolution amplitude was 0.12 mm. The convolution amplitude is one of the dimensions of the convoluted wall 34 that govern the conversion of axial load into radial expansion.

[0237] In the prototype valve cores the normal stroke 114 of the poppet 40 was 0.6 mm. This is the only value of the normal stroke 114 that has been built, and no range is stated for it.

[0238] In some embodiments the over-travel distance, which is the distance the poppet 40 moves beyond the normal stroke 114, is from 0.4 mm to 0.9 mm. In the prototype valve cores the over-travel distance was 0.5 mm. The over-travel distance is the travel during which the purge shoulder 44 bears on the seat ring 30 and applies the axial load.

[0239] In some embodiments the purge gap width, which is the size of the purge gap 60 when fully open, is from 0.10 mm to 0.25 mm. In the prototype valve cores the purge gap width was 0.15 mm. These values are stated as radial values. The radial expansion that opens the purge gap 60 arises from flattening of the convolution 36 under the axial load, as described under "Principal embodiment: operation".

[0240] In some embodiments the crack pressure 110 is from 0.8 bar to 1.5 bar. In the prototype valve cores the crack pressure 110 was 1.0 bar. The crack pressure 110 is the pressure differential at which the poppet 40 lifts off

the sealing edge 32 against the first spring stage 52.

[0241] In some embodiments the over-travel differential 112 is from 2.5 bar to 5.0 bar. In the prototype valve cores the over-travel differential 112 was 3.5 bar. The over-travel differential 112 is the pressure differential that overcomes the second spring stage 54 and produces the over-travel 116. Throughout the ranges stated for the crack pressure 110 and for the over-travel differential 112, the over-travel differential 112 is greater than the crack pressure 110, so that the poppet 40 opens through the normal stroke 114 at a lower pressure differential than that at which the purge shoulder 44 reaches the seat ring 30.

[0242] Narrower ranges are stated in the following paragraphs for each parameter that has a stated range. Each narrower range lies within the corresponding range stated above and contains the prototype value. The narrower ranges are fallback positions within the stated ranges. No valve core having a value other than the prototype value has been built or tested.

[0243] In some embodiments the seat ring inner diameter is from 2.1 mm to 2.5 mm. In some embodiments the seat ring inner diameter is from 2.2 mm to 2.4 mm.

[0244] In some embodiments the wall thickness is from 0.17 mm to 0.25 mm. In some embodiments the wall thickness is from 0.18 mm to 0.22 mm.

[0245] In some embodiments the free length is from 2.7 mm to 3.5 mm. In some embodiments the free length is from 2.8 mm to 3.2 mm.

[0246] In some embodiments the convolution amplitude is from 0.10 mm to 0.16 mm. In some embodiments the convolution amplitude is from 0.11 mm to 0.13 mm.

[0247] In some embodiments the over-travel distance is from 0.45 mm to 0.70 mm. In some embodiments the over-travel distance is from 0.45 mm to 0.55 mm.

[0248] In some embodiments the purge gap width is from 0.12 mm to 0.20 mm. In some embodiments the purge gap width is from 0.13 mm to 0.17 mm.

[0249] In some embodiments the crack pressure 110 is from 0.9 bar to 1.3 bar. In some embodiments the crack pressure 110 is from 0.9 bar to 1.1 bar.

[0250] In some embodiments the over-travel differential 112 is from 3.0 bar to 4.5 bar. In some embodiments the over-travel differential 112 is from 3.0 bar to 4.0 bar.

[0251] For each parameter, any lower limit stated in this section may be combined with any higher upper limit stated in this section for the same parameter, and the prototype value may serve as either limit.

[0252] Certain parameters have a single stated value and no stated range. The interference fit by which the seat ring 30 of the prototype valve cores was retained in the core body 20 was about 0.03 mm on diameter. The NiTi alloy of the seat ring 30 had a nominal composition of 55.8 wt% Ni. The shape-setting of the seat ring 30 was carried out at a temperature of 500 °C for a period of 10 minutes.

[0253] In some embodiments the parameters of the valve core 5 are defined by their function. In such embodiments the dimensions of the seat ring 30, the over-travel distance and the axial load applied by the purge shoulder 44 are such that the radial expansion of the seat ring 30 in the over-travel 116 is sufficient to open the purge gap 60, and the second spring stage 54 is such that the over-travel differential 112 is greater than the crack pressure 110.

Manufacture and assembly

[0254] FIG. 7 shows the sequence by which the seat ring 30 of the prototype valve cores was made and fitted: a laser cutting step 510, a shape-setting step 512, a quenching step 514, an electropolishing step 516 and a press-fitting step 518.

[0255] In the laser cutting step 510, the seat ring 30 is cut from a tube of NiTi alloy to the profile of the convoluted wall 34, the tube being held on a mandrel. The tube used for the prototype valve cores had an outside diameter of 2.7 mm and a wall of 0.20 mm.

[0256] In some embodiments the convolution 36 is formed in the shape-setting step 512, a fixture pressing the plain wall of the tube into one circumferential convolution or two, and the laser cutting step 510 trims the ends of the tube. In other embodiments the laser cutting step 510 cuts a pattern in the wall, as described under "Forms of the radially deflectable element".

[0257] In the shape-setting step 512, the seat ring 30 is held in a fixture that holds the convolution geometry, and is heated while so held. For the prototype valve cores the shape-setting was at a temperature of 500 °C for a period of 10 minutes. The shape-setting fixes in the NiTi alloy the convolution geometry held by the fixture, so that the convoluted wall 34 has that geometry at rest and returns to it on shape recovery.

[0258] In the quenching step 514, the seat ring 30 is cooled rapidly after the shape-setting. The seat rings of the prototype valve cores were quenched in water.

[0259] In the electropolishing step 516, the oxide layer and the heat-affected edge of the seat ring 30 are removed. Electropolishing is described for a seat ring 30 formed from a metal such as superelastic NiTi alloy.

[0260] In some embodiments the oxide layer and the heat-affected edge are removed by chemical etching or by abrasive-flow finishing in place of electropolishing.

[0261] In the press-fitting step 518, the seat ring 30 is pressed into the core body 20 with an interference fit and seated against the locating shoulder 24. In the prototype valve cores the interference fit was about 0.03 mm on diameter. Other ways of retaining the seat ring 30 are described under "Purge gap arrangements and retention of the seat ring".

[0262] The core body 20 and the poppet 40 are made by ordinary turning. No special process is used for them.

[0263] In some embodiments the polymer seat ring 130 is turned from rod, for example from rod of PEEK. In such embodiments the wall of the polymer seat ring 130 is supported during cutting, a wall thickness of 0.20 mm being thin for a turned polymer part. The shape-setting step 512 and the electropolishing step 516 are steps for a seat ring 30 of metal and are not used for the polymer seat ring 130.

[0264] In some embodiments the radially deflectable element is a slotted sleeve 170 that is stamped, deep-drawn or rolled from strip of spring-temper metal. No shape-setting is used for such a slotted sleeve 170.

[0265] In some embodiments the convolution amplitude is controlled at the fixture of the shape-setting step 512, so that the convolution amplitude of each seat ring 30 lies within a chosen part of the stated range. This control guards against a convolution amplitude too small to give the radial expansion that opens the purge gap 60.

[0266] In some embodiments a tolerance band is set for the convolution amplitude, and the assembled valve core 5 is checked with a gauge, before it is supplied, to confirm that the purge gap 60 is closed at rest. This check guards against a convolution amplitude so large that the seat ring 30 strains the interference fit or leaves the purge gap 60 open at rest.

[0267] In some embodiments the interference fit is held within a controlled range, and each valve core 5 is tested to confirm that the purge gap 60 opens at the intended over-travel differential 112. This control guards against an interference fit so tight that it restricts the radial expansion of the seat ring 30.

[0268] In some embodiments a batch of seat rings is verified after the shape-setting step 512 and before the press-fitting step 518, for example by a check of the transformation temperature. This verification guards against a seat ring 30 whose shape-setting was incomplete.

[0269] In some embodiments the electropolishing step 516 is carried out within limits of time and of current, and the wall thickness is checked in process, so that the convoluted wall 34 keeps its wall thickness and the profile of the convolution 36.

[0270] In some embodiments a minimum duration of the electropolishing step 516, or a surface finish to be reached by it, is specified, so that the oxide layer and the heat-affected edge are removed from the sealing edge 32.

Use: cleaning during an inflation cycle

[0271] FIG. 6 shows, in four panels, the states through which the valve core 5 passes in one inflation cycle: at rest 500, normal stroke 502, over-travel 504 and return 506. Each panel shows the poppet 40 with the sealing face 42 and the purge shoulder 44, the seat ring 30 and the purge gap 60. The behaviour of the valve core 5 in each state is described under "Principal embodiment: operation". The present section describes the inflation cycle as a method of cleaning the tyre valve 1.

[0272] The method starts with the valve core 5 at rest 500: the sealing face 42 is seated on the sealing edge 32 and the purge gap 60 is closed. A pump head 90 is coupled to the tyre valve 1 and applies a pressure differential across the poppet 40. When the pressure differential exceeds the crack pressure 110, the poppet 40 lifts through the normal stroke 114, as shown in the normal stroke panel 502, and air flows through the main passage 26 into the tyre chamber 80. The purge gap 60 remains closed.

[0273] When the pressure differential exceeds the over-travel differential 112, the poppet 40 moves through the over-travel 116, as shown in the over-travel panel 504. The purge shoulder 44 bears on the seat ring 30, the seat ring 30 undergoes radial expansion and the purge gap 60 opens. Air at pump pressure flows through the purge gap 60, alongside the air flowing through the main passage 26, across surfaces of the valve core 5 on which sealant film accumulates. This air purge carries sealant film into the tyre chamber 80.

[0274] When the pump head 90 is removed, the pressure differential collapses and the poppet 40 returns, as shown in the return panel 506. The axial load on the seat ring 30 is removed and the seat ring 30 undergoes shape recovery, which closes the purge gap 60. In embodiments in which the seat ring 30 is superelastic, the shape recovery also produces the snap-back wipe, as described under "Principal embodiment: operation".

[0275] The cleaning is part of the inflation cycle. The user couples the pump head 90, inflates the tyre chamber 80 and removes the pump head 90. The valve core 5 stays in place in the valve stem 10 and is cleaned without a tool. The cleaning takes place in each inflation cycle in which the pressure differential exceeds the over-travel differential 112.

[0276] The method can be carried out with any pump head 90 that applies a pressure differential exceeding the over-travel differential 112. In Example 1 the pump head 90 was that of a floor pump, the tyre chamber 80 was inflated to 6 bar, and over-travel was confirmed on every inflation of a prototype valve core. The rate at which the floor pump delivered air was not recorded.

[0277] Inflation from a CO■ cartridge has not been tested. A CO■ cartridge inflator chills the tyre valve 1 to well below 0 °C for a few seconds. If the temperature of a seat ring 30 of NiTi alloy falls below the martensite start temperature of the NiTi alloy, the seat ring 30 loses its superelastic recovery until it warms.

[0278] In some embodiments the valve core 5 is supplied with instructions for use that recommend inflation with a floor pump. The choice of the austenite finish temperature for cold conditions is described under "Materials and shape recovery".

[0279] The sealant with which the method has been tested is a latex emulsion sealant. Sealant of this kind typically contains ammonia and suspended particles. No ammonia-free sealant has been tested.

Examples and data

[0280] Example 1 is an actual example: the valve cores were built and the results were measured. Examples 2, 3 and 4 are prophetic examples: none of them has been built or tested, and no result is reported for any of them.

Example 1 (actual): valve core having a seat ring of superelastic NiTi alloy

[0281] Three prototype valve cores, designated P-1, P-2 and P-3, were built. Each was a valve core 5 as described under "Principal embodiment: structure", fitted in a Presta-pattern valve stem 10 of brass having an outer diameter of 6 mm. The core body 20 was turned from brass of grade CW614N. The poppet 40 was of brass and had a conical sealing face 42 and, below the sealing face 42, a purge shoulder 44. The return spring 50 was of stainless music wire and had a two-rate characteristic. The seat ring 30 was of superelastic NiTi alloy of nominal composition 55.8 wt% Ni, with an austenite finish temperature specified as below 5 °C, and had a convoluted wall 34.

[0282] The prototype values were: seat ring inner diameter 2.3 mm; wall thickness 0.20 mm; free length 3.0 mm; convolution amplitude 0.12 mm; normal stroke 0.6 mm; over-travel distance 0.5 mm; purge gap width, fully open, 0.15 mm, stated as a radial value; crack pressure 1.0 bar; over-travel differential 3.5 bar.

[0283] The seat rings were made and fitted by the steps 510 to 518 described under "Manufacture and assembly", with shape-setting at a temperature of 500 °C for a period of 10 minutes, quenching in water, electropolishing, and an interference fit of about 0.03 mm on diameter.

[0284] Three conventional valve cores, designated C-1, C-2 and C-3, were tested for comparison. They were standard Presta-pattern valve cores of a single manufacturer.

[0285] Each of the six valve cores was fitted to a tyre chamber 80 in the form of a bicycle inner tube of 700×28 size. The inner tubes were identical and each was charged with a volume of 60 ml of latex emulsion sealant. One inflation cycle of the test consisted of inflating the tyre chamber 80 to 6 bar with a floor pump, holding for a period of 24 hours at a temperature of 22 °C, and deflating fully through the tyre valve 1. The inflation cycle was then repeated. The test is recorded as carried out from 12 August to 20 August of 2026.

[0286] A valve core was recorded as having failed when either of two conditions was met: a leak-down of more than 0.5 bar in a period of 24 hours, or the pump being unable to open the tyre valve 1 within 5 seconds of engagement of the pump head 90. The inflation time was measured from engagement of the pump head 90 until 6 bar was reached, with the same pump and the same operator throughout.

[0287] The results are set out in the following table. In the column of inflation times at the last inflation cycle, the number in brackets is the inflation cycle at which that inflation time was measured. It belongs to that inflation time only, and not to the other values in the same row.

Core	Type	Cycles to first failure	Failure mode	Inflation time, cycle 1 (s)	Inflation time, last cycle (s)	Leak-down at last cycle (bar/24 h)
C-1	Conventional	14	blocked (pump could not open)	41	68 (cycle no. 13)	0.1
C-2	Conventional	12	leak-down 0.7 bar	40	44 (cycle no. 11)	0.7
C-3	Conventional	18	blocked	42	75 (cycle no. 17)	0.2
P-1	Prototype (seat ring of NiTi alloy)	none up to 40	—	43	47 (cycle no. 40)	0.1
P-2	Prototype (seat ring of NiTi alloy)	none up to 40	—	41	46 (cycle no. 40)	0.1
P-3	Prototype (seat ring of NiTi alloy)	none up to 40	—	44	50 (cycle no. 40)	0.2

[0288] The three conventional valve cores failed at cycle counts of 14, 12 and 18 respectively, two by blocking and one by leak-down. Each of the three prototype valve cores completed a total of 40 inflation cycles with no failure, at which point the test was stopped. For the prototype valve cores the mean inflation time at the last inflation cycle was within 15% of the mean inflation time at the first inflation cycle. The leak-down of the prototype valve cores at the last inflation cycle was 0.1 bar, 0.1 bar and 0.2 bar in a period of 24 hours, against the failure criterion of 0.5 bar.

[0289] After the last inflation cycle the seat rings of the prototype valve cores were inspected under a stereo microscope. Cured sealant film was present on the outer surface of the seat ring 30 in the region of the purge gap 60. The sealing edge 32 and the sealing face 42 of the poppet 40 were clean. The surfaces on which the poppet 40 seals were therefore free of sealant film after more inflation cycles than those at which the conventional valve cores had failed by blocking or by leak-down.

[0290] Over-travel was confirmed on every inflation of a prototype valve core by an audible snap on removal of the pump head 90.

[0291] The evidence of Example 1 has the following limits. The test covered a total of 40 inflation cycles, one sealant, one floor pump, one temperature, one size of tyre chamber 80 and three valve cores of each type. No fatigue test and no cold test was carried out. The comparison was between the prototype valve cores and conventional valve cores. No valve core having the air purge alone or the snap-back wipe alone was tested, so the test does not show the separate contribution of each cleaning action. The behaviour of the seat ring 30 beyond the inflation cycles tested, including whether the purge gap 60 remains closed at rest after longer service, has not been established, and no statement is made as to service life.

Prophetic example 2: polymer seat ring

[0292] This example has not been built or tested. The valve core 5 would be as in Example 1, except that the polymer seat ring 130, of PEEK, would take the place of the seat ring 30 of NiTi alloy. The polymer seat ring 130 would have a sealing edge 132 and a convoluted wall 134 of the same geometry as in Example 1, with a seat ring inner diameter of 2.3 mm, a wall thickness of 0.20 mm, a free length of 3.0 mm and a convolution amplitude of 0.12 mm. No value is given for the radial expansion or for the purge gap width that these dimensions would give in PEEK.

[0293] The polymer seat ring 130 would be turned from rod as described under "Manufacture and assembly" and press-fitted against the locating shoulder 24. In use the poppet 40 would lift through the normal stroke 114 and, above the over-travel differential 112, would reach the over-travel 116 as in Example 1. The polymer seat ring 130 has some radial compliance. Its shape recovery on removal of the pump head 90 would be slower, and of lower force, than that of the seat ring 30 of NiTi alloy.

[0294] The air purge is expected to operate as in Example 1, because it depends on the opening of the purge gap 60 and not on the material of the seat ring. The snap-back wipe is expected to be weaker than in Example 1.

Prophetic example 3: two-piece seat

[0295] This example has not been built or tested. The valve core 5 would have a two-piece seat in place of the seat ring 30 of Example 1. A rigid seat 140 of brass would provide the sealing edge 142 on which the sealing face 42 of the poppet 40 seats. A wiper ring 144, being a thin ring of stainless steel of spring character, would lie below the rigid seat 140. The poppet 40, the return spring 50 and the remaining components would be as in Example 1. No dimensions are given for the rigid seat 140 or for the wiper ring 144.

[0296] At rest the seal would be made on the rigid seat 140 and would not depend on the wiper ring 144. During the over-travel 116 the purge shoulder 44 would deflect the wiper ring 144 outward. On removal of the pump head 90 the wiper ring 144 would undergo shape recovery and would wipe on its return.

[0297] Sealing on the rigid seat 140 is expected to be more predictable than sealing on an element that also deflects, and the two-piece seat may be easier to make. The two-piece seat would have one more part and a greater axial length than the seat ring 30 of Example 1.

Prophetic example 4: thermal variant

[0298] This example has not been built or tested, and there are no data for it. The seat ring would be the thermal variant 150, being a seat ring of NiTi alloy trained with two-way shape memory and having an austenite finish temperature of about 35 °C to 40 °C. Warming the valve core 5, for example by hand, would take the thermal variant 150 through that temperature range, whereupon the thermal variant 150 would change shape and open the purge gap 60 without a pump. On cooling, the thermal variant 150 would return to its earlier shape. Opening by warming would not use the over-travel 116 of the poppet 40. The thermal variant 150 is further described under "Materials and shape recovery".

Claim-support statements

[0299] The following numbered statements set out combinations of the features described above. Each statement is to be read with the definitions given under "Definitions and interpretation". A statement that refers to an earlier statement includes all the features of that earlier statement.

[0300] Statement 1.1. A valve core for a tyre valve, the valve core comprising: a core body; a sealing edge; a poppet having a sealing face that closes against the sealing edge at rest; a radially deflectable element; and a load-applying surface moved by the poppet; wherein the poppet is movable away from the sealing edge through a normal stroke, in which a main passage past the sealing edge is open, and through an over-travel beyond the normal stroke; wherein, in the over-travel, the load-applying surface bears on the radially deflectable element and applies an axial load that produces a radial expansion of the radially deflectable element, the radial expansion opening a purge gap; wherein the purge gap is a secondary flow path that is distinct from the main passage and is closed at rest; and wherein the radially deflectable element undergoes shape recovery when the axial load is removed, the shape recovery closing the purge gap.

[0301] Statement 1.2. The valve core of statement 1.1, wherein the load-applying surface is spaced from the radially deflectable element through the normal stroke and bears on the radially deflectable element only in the over-travel.

[0302] Statement 1.3. The valve core of statement 1.1 or 1.2, wherein the poppet lifts off the sealing edge when a pressure differential across the poppet exceeds a crack pressure, and moves through the over-travel when the pressure differential exceeds an over-travel differential that is greater than the crack pressure.

[0303] Statement 1.4. The valve core of statement 1.3, further comprising a return bias that biases the poppet towards the sealing edge and has a two-rate characteristic, the two-rate characteristic having a first spring stage that the crack pressure overcomes and a second spring stage that the over-travel differential overcomes.

[0304] Statement 1.5. The valve core of statement 1.4, wherein the return bias is a return spring.

[0305] Statement 1.6. The valve core of any of statements 1.1 to 1.5, wherein the radially deflectable element is a seat ring retained in the core body, and the sealing edge is an edge of the seat ring, so that one part both seals against the poppet and opens the purge gap.

[0306] Statement 1.7. The valve core of statement 1.6, wherein the seat ring has a load-converting geometry that converts the axial load into the radial expansion.

[0307] Statement 1.8. The valve core of statement 1.7, wherein the load-converting geometry is a convoluted wall having at least one convolution, the axial load flattening the convolution.

[0308] Statement 1.9. The valve core of statement 1.8, wherein the convoluted wall has one convolution or two convolutions.

[0309] Statement 1.10. The valve core of any of statements 1.1 to 1.9, wherein the purge gap is bounded by an outer surface of the radially deflectable element and by a bore surrounding the radially deflectable element.

[0310] Statement 1.11. The valve core of statement 1.10, wherein the bore is a core body bore of the core body.

[0311] Statement 1.12. The valve core of statement 1.10, wherein the bore is a bore of a valve stem in which the valve core is fitted.

[0312] Statement 1.13. The valve core of any of statements 1.10 to 1.12, wherein the purge gap is annular.

[0313] Statement 1.14. The valve core of any of statements 1.6 to 1.9, wherein the purge gap comprises a flow path through a wall of the seat ring.

[0314] Statement 1.15. The valve core of any of statements 1.1 to 1.14, wherein, when the purge gap is open, air at pump pressure flows through the purge gap, across a surface of the valve core on which sealant film accumulates, and into a tyre chamber, as an air purge.

[0315] Statement 1.16. The valve core of any of statements 1.1 to 1.15, wherein the load-applying surface is a purge shoulder of the poppet, located below the sealing face.

[0316] Statement 1.17. The valve core of any of statements 1.6 to 1.16, wherein the seat ring is of a superelastic material.

[0317] Statement 1.18. The valve core of statement 1.17, wherein the superelastic material is a NiTi alloy.

[0318] Statement 1.19. The valve core of statement 1.18, wherein the NiTi alloy has an austenite finish temperature below 5 °C.

[0319] Statement 1.20. The valve core of any of statements 1.17 to 1.19, wherein, on the shape recovery, the seat ring moves across the sealing face of the poppet in a snap-back wipe.

[0320] Statement 1.21. The valve core of any of statements 1.6 to 1.16, wherein the seat ring is a polymer seat ring, for example of PEEK.

[0321] Statement 1.22. The valve core of any of statements 1.6 to 1.21, wherein the seat ring is retained in the core body by an interference fit and is seated against a locating shoulder of the core body.

[0322] Statement 1.23. The valve core of any of statements 1.6 to 1.22, having one or more of: a seat ring inner diameter from 2.0 mm to 2.6 mm; a wall thickness from 0.15 mm to 0.30 mm; a free length from 2.5 mm to 4.0 mm; a convolution amplitude from 0.08 mm to 0.20 mm; an over-travel distance from 0.4 mm to 0.9 mm; a purge gap width from 0.10 mm to 0.25 mm; a crack pressure from 0.8 bar to 1.5 bar; and an over-travel differential from 2.5 bar to 5.0 bar.

[0323] Statement 1.24. The valve core of statement 1.23, having a seat ring inner diameter of 2.3 mm, a wall thickness of 0.20 mm, a free length of 3.0 mm, a convolution amplitude of 0.12 mm, a normal stroke of 0.6 mm, an over-travel distance of 0.5 mm, a purge gap width of 0.15 mm, a crack pressure of 1.0 bar and an over-travel differential of 3.5 bar.

[0324] Statement 1.25. The valve core of any of statements 1.1 to 1.24, having one or more of: a core body of brass or of stainless steel; a sealing face of hard-anodised aluminium; and a lock nut on the poppet.

[0325] Statement 2.1. A tyre valve comprising a valve stem and the valve core of any of statements 1.1 to 1.25, the valve core being received in the valve stem.

[0326] Statement 2.2. The tyre valve of statement 2.1, wherein the valve core is a removable threaded valve core.

[0327] Statement 2.3. The tyre valve of statement 2.1, wherein the valve core is a valve core integral with the valve stem.

[0328] Statement 2.4. The tyre valve of statement 2.1 or 2.2, being a Presta-pattern tyre valve.

[0329] Statement 2.5. A tyre chamber fitted with the tyre valve of any of statements 2.1 to 2.4, the tyre chamber containing sealant.

[0330] Statement 2.6. The tyre chamber of statement 2.5, being a bicycle inner tube, or a tubeless tyre and rim cavity.

[0331] Statement 3.1. A method of cleaning a tyre valve during an inflation cycle, the tyre valve having the valve core of any of statements 1.1 to 1.25, the method comprising: coupling a pump head to the tyre valve; applying with the pump head a pressure differential that moves the poppet through the normal stroke, so that air flows through the main passage into a tyre chamber; applying with the pump head a pressure differential that moves the poppet through the over-travel, so that the load-applying surface bears on the radially deflectable element and the purge gap opens; passing air through the open purge gap into the tyre chamber; and removing the pump head, whereupon the radially deflectable element undergoes shape recovery and the purge gap closes.

[0332] Statement 3.2. The method of statement 3.1, wherein the air passing through the open purge gap carries sealant film into the tyre chamber.

[0333] Statement 3.3. The method of statement 3.1 or 3.2, wherein the shape recovery produces a snap-back wipe.

[0334] Statement 3.4. The method of any of statements 3.1 to 3.3, carried out with the valve core in place in a valve stem of the tyre valve.

[0335] Statement 3.5. The method of any of statements 3.1 to 3.4, wherein the pump head is the pump head of a floor pump.

[0336] Statement 4.1. A seat ring for a valve core of a tyre valve, the seat ring being tubular and comprising: a sealing edge at one end, for a sealing face of a poppet of the valve core; and a wall having a load-converting geometry; wherein an axial load applied to the seat ring produces, through the load-converting geometry, a radial expansion of the wall; wherein, when the seat ring is installed in the valve core, the radial expansion opens a purge gap bounded by an outer surface of the seat ring and by a bore surrounding the seat ring; and wherein the seat ring undergoes shape recovery when the axial load is removed.

[0337] Statement 4.2. The seat ring of statement 4.1, wherein the load-converting geometry is a convoluted wall having at least one convolution.

[0338] Statement 4.3. The seat ring of statement 4.1 or 4.2, being of superelastic NiTi alloy having an austenite finish temperature below 5 °C.

[0339] Statement 4.4. The seat ring of statement 4.1 or 4.2, being a polymer seat ring.

[0340] Statement 4.5. The seat ring of any of statements 4.1 to 4.4, having one or more of: a seat ring inner diameter from 2.0 mm to 2.6 mm; a wall thickness from 0.15 mm to 0.30 mm; a free length from 2.5 mm to 4.0 mm; and a convolution amplitude from 0.08 mm to 0.20 mm.

[0341] Statement 4.6. The seat ring of any of statements 4.1 to 4.5, supplied as a replacement part for a valve core.

[0342] Statement 5.1. A method of manufacturing a seat ring for a valve core of a tyre valve, the method comprising: laser cutting the seat ring from a tube of NiTi alloy; shape-setting the seat ring in a fixture that holds a convolution geometry of a convoluted wall of the seat ring; quenching the seat ring; and electropolishing the seat ring.

[0343] Statement 5.2. The method of statement 5.1, wherein the shape-setting is at a temperature of 500 °C for a period of 10 minutes, and the quenching is in water.

[0344] Statement 5.3. The method of statement 5.1 or 5.2, wherein the electropolishing removes an oxide layer and a heat-affected edge from the seat ring.

[0345] Statement 5.4. The method of any of statements 5.1 to 5.3, further comprising press-fitting the seat ring into a core body of the valve core, with an interference fit, against a locating shoulder of the core body.

[0346] Statement 5.5. The method of statement 5.4, wherein the interference fit is about 0.03 mm on diameter.

[0347] Statement 5.6. A seat ring obtainable by the method of any of statements 5.1 to 5.3.

[0348] Statement 6.1. A replacement valve core, being the valve core of any of statements 1.1 to 1.25 in the form of a removable threaded valve core supplied separately from a valve stem.

[0349] Statement 7.1. A valve core for a tyre valve, the valve core comprising: a core body; a two-piece seat having a rigid seat that provides a sealing edge and a wiper ring below the rigid seat; and a poppet having a sealing face that closes against the sealing edge at rest, and a purge shoulder; wherein the poppet is movable away from the sealing edge through a normal stroke and through an over-travel beyond the normal stroke; wherein, in the over-travel, the purge shoulder deflects the wiper ring outward; and wherein the wiper ring undergoes shape recovery, and wipes on its return, when the purge shoulder withdraws.

[0350] Statement 7.2. The valve core of statement 7.1, wherein the rigid seat is of brass and the wiper ring is of stainless steel.

Closing interpretation

[0351] The embodiments and examples described above are examples of the valve core, the tyre valve, the seat ring and the methods. A feature described with the words "in some embodiments" is optional. A feature described for one embodiment may be used with another embodiment in the combinations set out under "Combinations".

[0352] The figures are schematic. The values stated for the prototype valve cores are the values of one built configuration and do not limit the embodiments for which a range, or no value, is stated.

List of Figures

Fig. 1 — simulated figure.

Claims

- A valve core (5) for a tyre valve (1) that admits air into a tyre chamber (80), the valve core comprising:

a core body (20); a sealing edge (32, 132, 142); a poppet (40) having a sealing face (42) that seats on the sealing edge when the poppet is in a closed position, the poppet being movable from the closed position through a normal stroke (114), in which a main passage (26) past the sealing edge is open, and through an over-travel (116) beyond the normal stroke; a radially deflectable element; and a load-applying surface that is moved by the poppet; wherein the load-applying surface is spaced from the radially deflectable element throughout the normal stroke; wherein, in the over-travel, the load-applying surface bears on the radially deflectable element and applies an axial load that produces a radial expansion of the radially deflectable element, the radial expansion opening a purge gap (60, 146); wherein the purge gap is a secondary flow path that is distinct from the main passage and that is closed at rest and throughout the normal stroke; and wherein the radially deflectable element undergoes shape recovery, which closes the purge gap, when the axial load is removed.

- A method of cleaning a tyre valve (1) during an inflation cycle, the tyre valve having a valve core (5) that has a sealing edge (32), a poppet (40) having a sealing face (42) that is seated on the sealing edge in a closed position of the poppet, a radially deflectable element, and a load-applying surface that is moved by the poppet, the method comprising:

moving the poppet from the closed position through a normal stroke (114), so that air flows through a main passage (26) past the sealing edge into a tyre chamber (80) while a purge gap (60) remains closed; moving the poppet through an over-travel (116) beyond the normal stroke, so that the load-applying surface bears on the radially deflectable element and applies an axial load that produces a radial expansion of the radially deflectable element, the radial expansion opening the purge gap, the purge gap being a secondary flow path that is distinct from the main passage; passing air through the open purge gap into the tyre chamber; and returning the poppet to the closed position, so that the axial load is removed and the purge gap closes.

- A seat ring (30) for a valve core (5) of a tyre valve (1), the seat ring being tubular and comprising:

a sealing edge (32) at one end of the seat ring, for seating a sealing face (42) of a poppet (40) of the valve core; and a wall having a load-converting geometry; wherein an axial load, applied to the seat ring by a load-applying surface that is moved by the poppet in an over-travel (116) of the poppet beyond a normal stroke (114), produces through the load-converting geometry a radial expansion of the wall; wherein, when the seat ring is installed in the valve core, the radial expansion opens a purge gap (60) that is distinct from a main passage (26) of the valve core; and wherein the seat ring undergoes shape recovery, which closes the purge gap, when the axial load is removed.

- A method of manufacturing the seat ring (30) of claim 3, the seat ring being of a NiTi alloy, the method comprising shape-setting (512) the seat ring in a fixture that holds the load-converting geometry.
- The valve core of claim 1, wherein the poppet lifts off the sealing edge when a pressure differential across the poppet exceeds a crack pressure (110), and moves through the over-travel when the pressure differential exceeds an over-travel differential (112) that is greater than the crack pressure.
- The valve core of claim 5, further comprising a return bias that biases the poppet towards the closed position and that has a two-rate characteristic, the two-rate characteristic having a first spring stage (52), which the crack pressure overcomes, and a second spring stage (54), which the over-travel differential overcomes.
- The valve core of claim 6, wherein the return bias is a return spring (50).
- The valve core of claim 7, wherein the return spring is a single compression spring of variable pitch.
- The valve core of claim 6, wherein the return bias comprises a soft spring and a stiff spring arranged in series, or comprises nested springs that act in parallel.
- The valve core of claim 6, wherein the second spring stage is provided by an elastomeric bumper, which the poppet reaches at an end of the normal stroke, or by a disc spring, which is engaged at the end of the normal stroke.
- The valve core of any one of claims 5 to 10, wherein the crack pressure is from 0.8 bar to 1.5 bar and the over-travel differential is from 2.5 bar to 5.0 bar.
- The valve core of claim 11, wherein the crack pressure is 1.0 bar and the over-travel differential is 3.5 bar.
- The valve core of any one of claims 1 and 5 to 12, wherein the load-applying surface is a purge shoulder (44) formed on the poppet.
- The valve core of claim 13, wherein the purge shoulder is an annular surface of the poppet located below the sealing face.
- The valve core of any one of claims 1 and 5 to 12, wherein the load-applying surface is provided on a collar that is a separate part carried by the poppet.
- The valve core of any one of claims 1 and 5 to 15, wherein the purge gap is bounded by an outer surface of the radially deflectable element and by a bore surrounding the radially deflectable element.
- The valve core of claim 16, wherein the bore is a core body bore (22) of the core body.
- The valve core of claim 16 or claim 17, wherein the purge gap is annular.
- The valve core of any one of claims 16 to 18, wherein the purge gap has a purge gap width of from 0.10 mm to 0.25 mm.
- The valve core of any one of claims 1 and 5 to 19, wherein the radially deflectable element is a seat ring (30) retained in the core body, and the sealing edge is an edge of the seat ring.
- The valve core of claim 20, wherein the seat ring has a convoluted wall (34, 134) having at least one convolution (36), the convoluted wall converting the axial load into the radial expansion.
- The valve core of claim 21, wherein the axial load flattens the at least one convolution, the convoluted wall deforming by bending.
- The valve core of claim 21 or claim 22, wherein the convoluted wall has two convolutions (36, 38).

- The valve core of claim 21 or claim 22, wherein the at least one convolution is a single outward bow of the convoluted wall.
- The valve core of any one of claims 20 to 24, wherein the seat ring is retained in the core body by an interference fit and is seated against a locating shoulder (24) of the core body.
- The valve core of claim 25, wherein the interference fit acts on a fit band (35) of the seat ring, and the seat ring has a free portion (37) which, on the radial expansion, expands into a core body bore relief (202) of the core body.
- The valve core of any one of claims 20 to 26, wherein the seat ring is of a superelastic material.
- The valve core of claim 27, wherein the superelastic material is a NiTi alloy.
- The valve core of claim 28, wherein the NiTi alloy has an austenite finish temperature below 5 °C.
- The valve core of any one of claims 20 to 26, wherein the seat ring is a polymer seat ring (130).
- The valve core of claim 30, wherein the polymer seat ring is of polyether ether ketone (PEEK).
- The valve core of any one of claims 20 to 31, wherein, on the shape recovery, the seat ring moves across the sealing face of the poppet in a snap-back wipe.
- The valve core of any one of claims 1 and 5 to 32, wherein the over-travel has an over-travel distance of from 0.4 mm to 0.9 mm.
- The valve core of claim 33, wherein the normal stroke is 0.6 mm and the over-travel distance is 0.5 mm.
- The valve core of any one of claims 1 and 5 to 34, wherein the core body is of stainless steel.
- The valve core of any one of claims 1 and 5 to 35, wherein the sealing face is of hard-anodised aluminium.
- The valve core of any one of claims 1 and 5 to 36, further comprising a lock nut (70) on the poppet, by which the poppet is closed manually.
- The valve core of any one of claims 1 and 5 to 37, wherein, when the purge gap is open, air at pump pressure flows through the purge gap, across a surface of the valve core on which sealant film accumulates, and into the tyre chamber, as an air purge.
- The valve core of claim 13, wherein the purge shoulder is conical.
- The valve core of claim 39, wherein the radially deflectable element is a slotted sleeve (170) having fingers (172) that the purge shoulder spreads in the over-travel.
- The valve core of claim 40, wherein the slotted sleeve is of a spring-tempered metal.
- The valve core of claim 39, wherein the radially deflectable element is a split seat ring (180) having an axial split (182) that the purge shoulder opens in the over-travel.
- The valve core of any one of claims 1 and 5 to 12, wherein the radially deflectable element is a tapered seat ring (190) having a taper (192), and the load-applying surface is a matching taper (194) of the poppet that wedges the tapered seat ring outwards in the over-travel.
- The valve core of any one of claims 1 and 5 to 15, wherein the radially deflectable element is a seat ring (30) having a convoluted wall (34), and the purge gap comprises wall slots (204) through the convoluted wall, the wall slots being closed at rest and open on the radial expansion.
- The valve core of claim 16, wherein the bore is a bore of a valve stem (10) in which the valve core is fitted.
- The valve core of any one of claims 1 and 5 to 44, being a removable threaded valve core, the core body having a thread by which the core body is screwed into a bore of a valve stem (10) of the tyre valve.
- A tyre valve (1) comprising a valve stem (10) and the valve core (5) of any one of claims 1 and 5 to 44, the valve core being received in a bore of the valve stem.
- The tyre valve of claim 47, wherein the valve core is a removable threaded valve core, the core body being screwed into the bore of the valve stem.

- The tyre valve of claim 47, wherein the valve core is a valve core integral with the valve stem (230), the core body being formed in the valve stem.
- The tyre valve of any one of claims 47 to 49, wherein the valve stem is fixed to the tyre chamber, and the tyre chamber contains sealant.
- The method of claim 2, wherein the poppet is moved through the normal stroke by a pressure differential across the poppet that exceeds a crack pressure (110), and is moved through the over-travel by the pressure differential when the pressure differential exceeds an over-travel differential (112) that is greater than the crack pressure.
- The method of claim 51, wherein the pressure differential is applied by a pump head (90) coupled to the tyre valve.
- The method of claim 51 or claim 52, wherein the crack pressure is from 0.8 bar to 1.5 bar and the over-travel differential is from 2.5 bar to 5.0 bar.
- The method of any one of claims 2 and 51 to 53, wherein the load-applying surface is a purge shoulder (44) formed on the poppet.
- The method of any one of claims 2 and 51 to 54, wherein, when the axial load is removed, the radially deflectable element undergoes shape recovery, which closes the purge gap.
- The method of claim 55, wherein the radially deflectable element is a seat ring (30) and the shape recovery produces a snap-back wipe, in which the seat ring moves across the sealing face of the poppet.
- The method of any one of claims 2 and 51 to 56, wherein the tyre chamber contains sealant, and the air passing through the open purge gap flows, as an air purge, across a surface of the valve core on which sealant film is present.
- The method of any one of claims 2 and 51 to 57, wherein the purge gap is bounded by an outer surface of the radially deflectable element and by a bore surrounding the radially deflectable element, the air passing through the open purge gap flowing over the outer surface.
- The method of any one of claims 2 and 51 to 58, carried out with the valve core in place in a valve stem (10) of the tyre valve.
- The method of any one of claims 2 and 51 to 59, wherein the method is repeated in each of a plurality of inflation cycles.
- The method of claim 2, wherein the valve core is the valve core of any one of claims 1 and 5 to 46.
- The seat ring of claim 3, wherein the load-converting geometry is a convoluted wall (34) having at least one convolution (36).
- The seat ring of claim 62, wherein the convoluted wall has two convolutions (36, 38).
- The seat ring of claim 62, wherein the at least one convolution is a single outward bow of the convoluted wall.
- The seat ring of any one of claims 3 and 62 to 64, wherein the purge gap is bounded by an outer surface of the seat ring and by a bore surrounding the seat ring.
- The seat ring of claim 65, wherein the outer surface comprises a fit band (35), for retention of the seat ring by an interference fit, and a free portion (37) that undergoes the radial expansion.
- The seat ring of any one of claims 3 and 62 to 64, wherein the purge gap comprises wall slots (204) through the wall, the wall slots being closed at rest and open on the radial expansion.
- The seat ring of any one of claims 3 and 62 to 67, having at least one of: a seat ring inner diameter of from 2.0 mm to 2.6 mm; a wall thickness of from 0.15 mm to 0.30 mm; a free length of from 2.5 mm to 4.0 mm; and a convolution amplitude of from 0.08 mm to 0.20 mm.
- The seat ring of any one of claims 3 and 62 to 68, having a seat ring inner diameter of 2.3 mm, a wall thickness of 0.20 mm, a free length of 3.0 mm and a convolution amplitude of 0.12 mm.
- The seat ring of any one of claims 3 and 62 to 69, wherein the seat ring is of a superelastic material.

- The seat ring of claim 70, wherein the superelastic material is a NiTi alloy.
- The seat ring of claim 71, wherein the NiTi alloy has an austenite finish temperature below 5 °C.
- The seat ring of claim 71 or claim 72, wherein the NiTi alloy has a nominal composition of 55.8 wt% nickel.
- The seat ring of any one of claims 3 and 62 to 69, wherein the seat ring is a polymer seat ring (130).
- The seat ring of claim 74, wherein the polymer seat ring is of polyether ether ketone (PEEK).
- The seat ring of any one of claims 3 and 62 to 75, wherein, on the shape recovery, the seat ring moves across the sealing face of the poppet in a snap-back wipe.
- A valve core (5) comprising a core body (20), a poppet (40) having a sealing face (42), and the seat ring (30) of any one of claims 3 and 62 to 76, the seat ring being retained in the core body.
- The method of claim 4, further comprising, before the shape-setting, laser cutting (510) the seat ring from a tube of the NiTi alloy, the tube being held on a mandrel.
- The method of claim 78, wherein the load-converting geometry is a convoluted wall (34), and the laser cutting cuts the seat ring to a profile of the convoluted wall.
- The method of claim 4 or claim 78, wherein the load-converting geometry is a convoluted wall (34) having at least one convolution (36), and the at least one convolution is formed in the shape-setting, the fixture pressing a plain wall of the seat ring into at least one circumferential wave.
- The method of any one of claims 4 and 78 to 80, wherein the shape-setting is carried out at a temperature of 500 °C for a period of 10 minutes.
- The method of any one of claims 4 and 78 to 81, further comprising quenching (514) the seat ring in water after the shape-setting.
- The method of claim 82, further comprising, after the quenching, electropolishing (516) the seat ring to remove an oxide layer and a heat-affected edge.
- The method of any one of claims 4 and 78 to 83, further comprising press-fitting (518) the seat ring into a core body (20) of the valve core, with an interference fit, against a locating shoulder (24) of the core body.
- The method of claim 84, wherein the interference fit is about 0.03 mm on diameter.
- The method of claim 4, wherein the seat ring is the seat ring of any one of claims 62 to 69.